# Theoretical Dataset Exploration

*`array_geometry_lab/`: 10 Geometry Families, Ghost/Vignetting Margins, Seed Variance*

Source: `sweep_results.csv` (6615 rows), `family_ranking.csv` (529 structurally-safe rows) – read live, not hardcoded

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# Preface: what this document is, and what it deliberately is not

This is a **working data-exploration document**, not a polished report and not a recommendation. Its only job is to make the actual shape of the theoretical dataset visible – margins, seed-to-seed variance, packing density – *before* any gated ranking or comparison mechanism is designed. Thresholds for a future safety/ranking gate should be chosen by looking at real distributions, not guessed in advance; this document is the looking step.

**This document does not draft or imply a final ranking, a gate threshold, or a "winner" anywhere.** Captions describe what is shown, not what should be concluded from it – that is explicitly deferred to a later, separate step. Where a pattern is visually obvious in a figure, it is named factually (e.g. "a hard margin cliff appears between $X$ and $Y$") without a recommendation attached.

**Family count note**: the underlying data contains 10 distinct `family` labels. `fibonacci_spiral` and `golden_angle` are, by construction, literally the same geometric pattern (documented identity in `generators/golden_angle.py`) – they are shown separately throughout this document because the data itself distinguishes them by name, but every figure for one is expected to be structurally identical to the other's.

**Every figure and table number in this document traces to a real row in `sweep_results.csv` or `family_ranking.csv`**, generated fresh by `generate_exploration_figures.py` – nothing here is hand-drawn, illustrative, or invented outside the actually-tested parameter ranges.

## Chapter 1

# Design Catalog

One representative layout per family: mid-range $N$ and mid-range primary spacing value, both drawn from parameter values *actually swept* (never invented), at $M = 4\text{D}$ ($a_{\text{hole}} = 600\,\mu\text{m}$, read live from `per_M_optimum_map.csv`). Hole circles are drawn to scale. For the two stochastic families, 3 independent seeds are shown side by side so seed-to-seed layout variation is visible directly, before Chapter 3 turns it into a number.

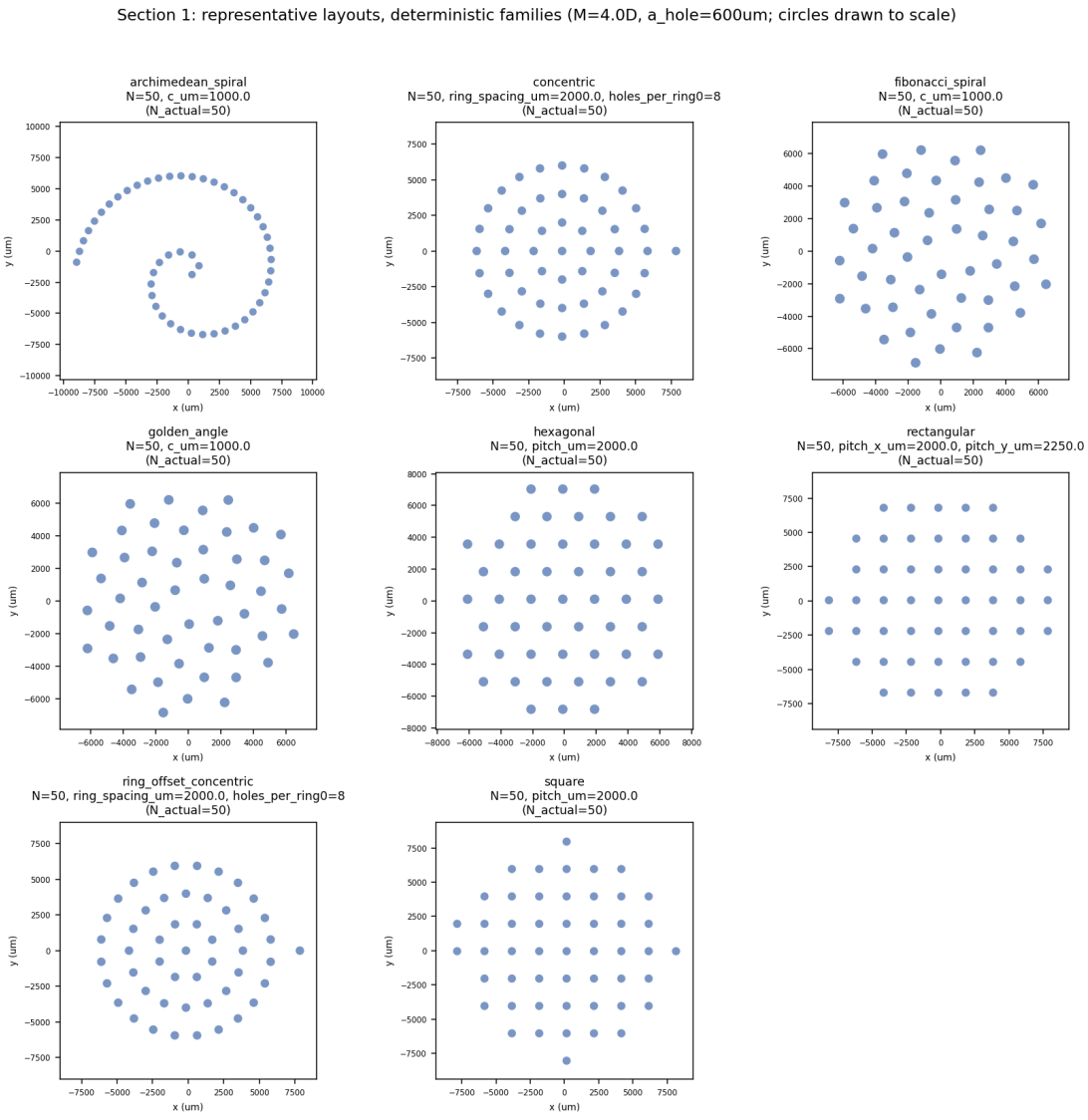



Figure 1.1: Representative layout, all 8 deterministic families, at their own mid-range tested $(N, \text{spacing})$.

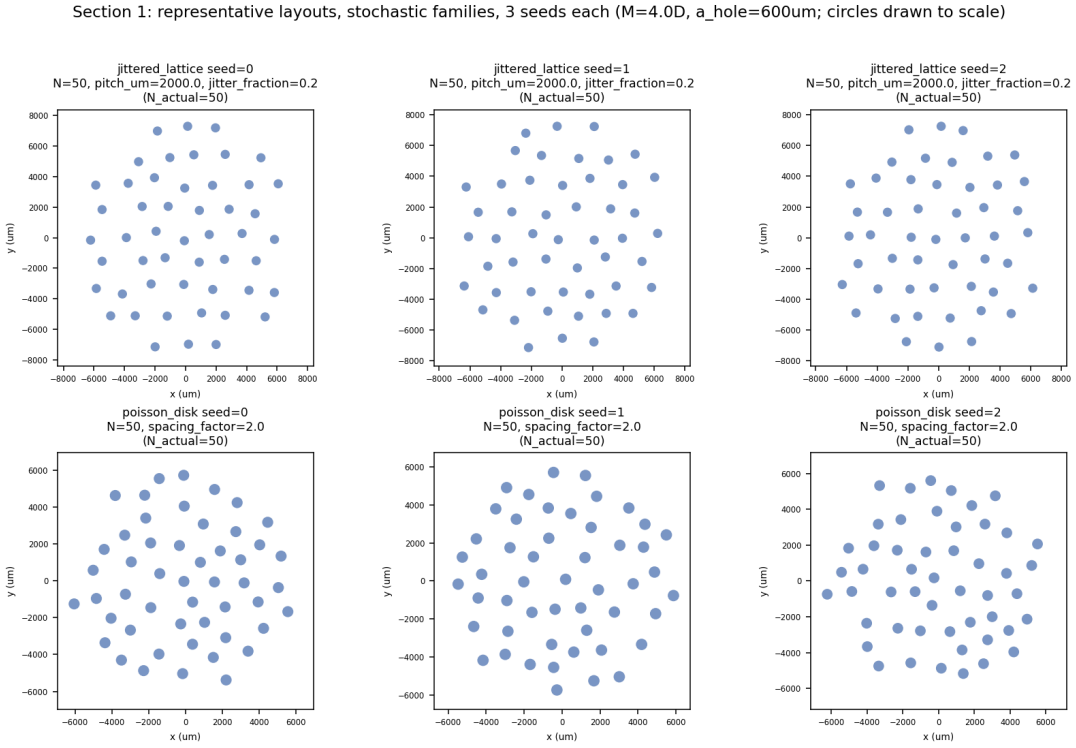



Figure 1.2: Representative layout, both stochastic families, 3 seeds each (same $N$/spacing point, seed 0/1/2) – the visible difference between panels in a row is seed-driven layout variation at fixed parameters.

Chapter 2

# Margin Heatmaps

For each family, four items, in this order: (1) a heatmap of `worst_case_ghost_margin` and `worst_case_vignetting_margin` over ($N$, primary spacing parameter), one small panel per tested $M$ (including `global`) – **every cell is now annotated with its own exact numeric value** (or `inf`), not color alone; (2) a sign-agreement map over the same axes, coloring each cell by whether the two margins' *signs* agree (both positive, both negative) or disagree, per $M$; (3) a line-plot view of the same data (margin vs. $N$, one line per primary spacing value, faceted by $M$) as a complementary graphical representation for reading trends across $N$ directly, which a color grid can obscure; (4) a raw number list – one row per tested ($N$, primary spacing, $M$) point, listing the exact `worst_case_ghost_margin`/`worst_case_vignetting_margin` values with no clipping or color encoding, for readers who want the literal numbers behind items (1)–(3). Where a family has a secondary spacing parameter (e.g. `holes_per_ring0`, `jitter_fraction`), it is held fixed at its own median tested value – stated in each heatmap's own caption/subtitle – so every heatmap's two axes are exactly ($N$, primary spacing), nothing folded in silently. For the two stochastic families, every cell (heatmap, line plot, and raw table alike) is the seed-aggregated worst-case value computed directly from `sweep_results.csv` over *all* tested seeds at that point – not only the seeds that happened to end up in `family_ranking.csv`'s safe-only subset, which would silently leave gaps at every unsafe point (a real gap in an earlier draft of this chapter, fixed by computing all four items from one shared grid-builder function that reads the full sweep, both here and in Chapter 3). Per-seed *distributional* detail (the spread across individual seeds, not just their worst-case) remains Chapter 3's job.

**Color/display convention, stated once**: heatmap color is clipped to $[-5, 5]$; a cell with $+\infty$ margin (zero active hole pairs under the pupil – a structural, not narrowly-numerical, safe result) is shown at the clipped color maximum but its printed cell text always reads `inf`, and the raw number list always reports the literal value – clipping is a display-only convention for the color scale, never applied to any number actually printed or listed.

## 2.1 archimedean_spiral

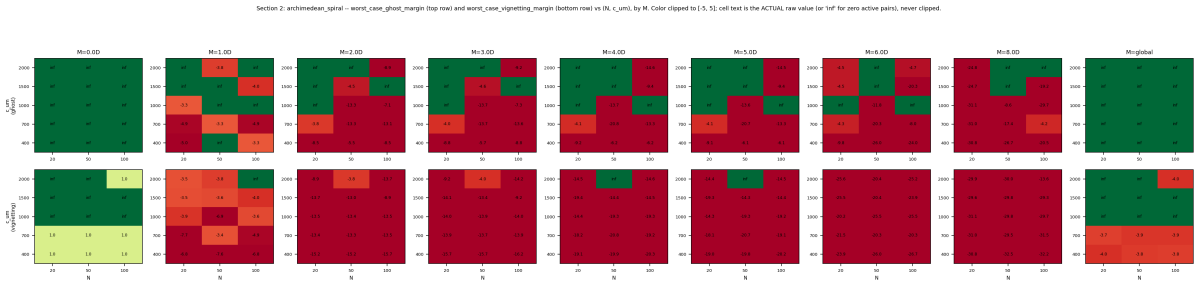



Figure 2.1: archimedean_spiral: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \text{c_um})$, by $M$. axes: $N \in \{20, 50, 100\}$.

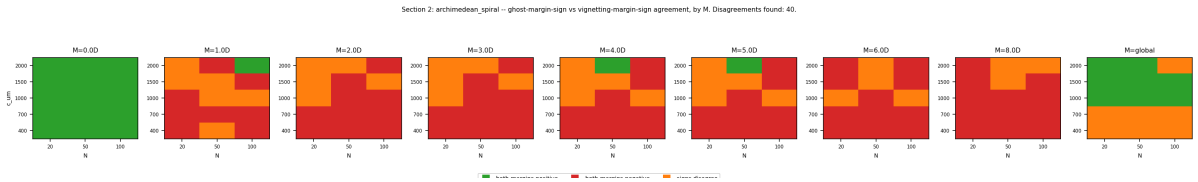



Figure 2.2: archimedean_spiral: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

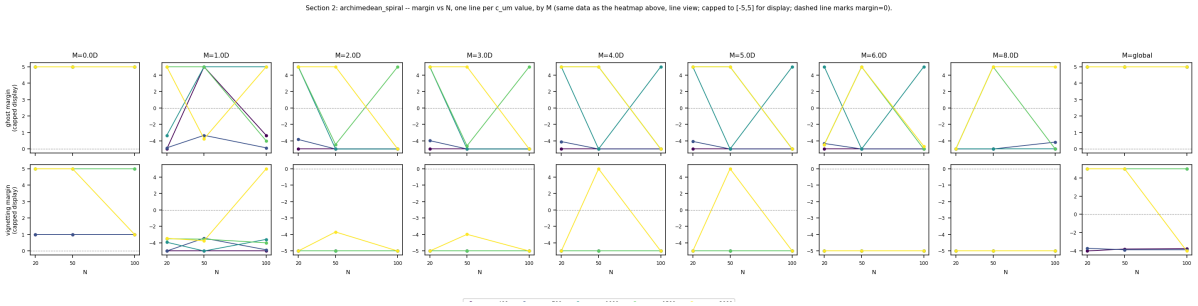



Figure 2.3: archimedean_spiral: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per c_um value) – a complementary graphical view for reading trends across $N$ directly.

### archimedean_spiral: raw number list

Table 2.1: archimedean_spiral: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, c_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 400 | 0D | inf | 1.000 |
| 20 | 400 | 1D | -4.996 | -6.762 |
| 20 | 400 | 2D | -8.487 | -15.237 |
| 20 | 400 | 3D | -8.768 | -15.717 |
| 20 | 400 | 4D | -9.186 | -19.128 |
| 20 | 400 | 5D | -9.143 | -19.042 |
| 20 | 400 | 6D | -9.800 | -23.865 |
| 20 | 400 | 8D | -30.798 | -30.798 |
| 20 | 400 | global | inf | -4.013 |
| 50 | 400 | 0D | inf | 1.000 |
| 50 | 400 | 1D | inf | -7.624 |
| 50 | 400 | 2D | -5.465 | -15.237 |
| 50 | 400 | 3D | -5.656 | -15.717 |
| 50 | 400 | 4D | -6.173 | -19.874 |
| 50 | 400 | 5D | -6.142 | -19.785 |
| 50 | 400 | 6D | -25.994 | -26.044 |
| 50 | 400 | 8D | -26.702 | -32.483 |
| 50 | 400 | global | inf | -3.808 |
| 100 | 400 | 0D | inf | 1.000 |
| 100 | 400 | 1D | -3.337 | -6.762 |
| 100 | 400 | 2D | -8.487 | -15.747 |
| 100 | 400 | 3D | -8.768 | -16.243 |
| 100 | 400 | 4D | -6.173 | -20.254 |
| 100 | 400 | 5D | -6.142 | -20.164 |
| 100 | 400 | 6D | -24.034 | -26.733 |
| 100 | 400 | 8D | -20.535 | -32.226 |
| 100 | 400 | global | inf | -3.770 |
| 20 | 700 | 0D | inf | 1.000 |
| 20 | 700 | 1D | -4.854 | -7.683 |
| 20 | 700 | 2D | -3.842 | -13.442 |
| 20 | 700 | 3D | -3.986 | -13.869 |
| 20 | 700 | 4D | -4.101 | -18.192 |
| 20 | 700 | 5D | -4.079 | -18.111 |
| 20 | 700 | 6D | -4.331 | -21.487 |
| 20 | 700 | 8D | -31.017 | -31.017 |
| 20 | 700 | global | inf | -3.729 |
| 50 | 700 | 0D | inf | 1.000 |
| 50 | 700 | 1D | -3.337 | -3.441 |
| 50 | 700 | 2D | -13.291 | -13.291 |
| 50 | 700 | 3D | -13.714 | -13.714 |
| 50 | 700 | 4D | -20.806 | -20.806 |

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| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 700 | 5D | -20.713 | -20.713 |
| 50 | 700 | 6D | -20.265 | -20.265 |
| 50 | 700 | 8D | -17.356 | -29.462 |
| 50 | 700 | global | inf | -3.888 |
| 100 | 700 | 0D | inf | 1.000 |
| 100 | 700 | 1D | -4.854 | -4.854 |
| 100 | 700 | 2D | -13.145 | -13.496 |
| 100 | 700 | 3D | -13.563 | -13.925 |
| 100 | 700 | 4D | -13.324 | -19.154 |
| 100 | 700 | 5D | -13.263 | -19.069 |
| 100 | 700 | 6D | -7.969 | -20.265 |
| 100 | 700 | 8D | -4.164 | -31.532 |
| 100 | 700 | global | inf | -3.888 |
| 20 | 1000 | 0D | inf | inf |
| 20 | 1000 | 1D | -3.337 | -3.934 |
| 20 | 1000 | 2D | inf | -13.547 |
| 20 | 1000 | 3D | inf | -13.977 |
| 20 | 1000 | 4D | inf | -14.393 |
| 20 | 1000 | 5D | inf | -14.327 |
| 20 | 1000 | 6D | inf | -20.212 |
| 20 | 1000 | 8D | -31.099 | -31.099 |
| 20 | 1000 | global | inf | inf |
| 50 | 1000 | 0D | inf | inf |
| 50 | 1000 | 1D | inf | -6.857 |
| 50 | 1000 | 2D | -13.285 | -13.430 |
| 50 | 1000 | 3D | -13.708 | -13.857 |
| 50 | 1000 | 4D | -13.661 | -19.347 |
| 50 | 1000 | 5D | -13.599 | -19.260 |
| 50 | 1000 | 6D | -11.770 | -25.505 |
| 50 | 1000 | 8D | -8.593 | -29.771 |
| 50 | 1000 | global | inf | inf |
| 100 | 1000 | 0D | inf | inf |
| 100 | 1000 | 1D | inf | -3.588 |
| 100 | 1000 | 2D | -7.060 | -13.546 |
| 100 | 1000 | 3D | -7.298 | -13.976 |
| 100 | 1000 | 4D | inf | -19.332 |
| 100 | 1000 | 5D | inf | -19.246 |
| 100 | 1000 | 6D | inf | -25.467 |
| 100 | 1000 | 8D | -29.705 | -29.705 |
| 100 | 1000 | global | inf | inf |
| 20 | 1500 | 0D | inf | inf |
| 20 | 1500 | 1D | inf | -3.515 |
| 20 | 1500 | 2D | inf | -13.655 |
| 20 | 1500 | 3D | inf | -14.088 |
| 20 | 1500 | 4D | inf | -19.411 |
| 20 | 1500 | 5D | inf | -19.324 |
| 20 | 1500 | 6D | -4.460 | -25.456 |
| 20 | 1500 | 8D | -24.713 | -29.565 |
| 20 | 1500 | global | inf | inf |

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| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1500 | 0D | inf | inf |
| 50 | 1500 | 1D | inf | -3.556 |
| 50 | 1500 | 2D | -4.455 | -13.024 |
| 50 | 1500 | 3D | -4.616 | -13.439 |
| 50 | 1500 | 4D | inf | -14.362 |
| 50 | 1500 | 5D | inf | -14.297 |
| 50 | 1500 | 6D | inf | -20.367 |
| 50 | 1500 | 8D | inf | -29.813 |
| 50 | 1500 | global | inf | inf |
| 100 | 1500 | 0D | inf | inf |
| 100 | 1500 | 1D | -4.005 | -4.005 |
| 100 | 1500 | 2D | inf | -8.862 |
| 100 | 1500 | 3D | inf | -9.154 |
| 100 | 1500 | 4D | -9.426 | -14.486 |
| 100 | 1500 | 5D | -9.382 | -14.420 |
| 100 | 1500 | 6D | -20.254 | -23.931 |
| 100 | 1500 | 8D | -19.189 | -29.253 |
| 100 | 1500 | global | inf | inf |
| 20 | 2000 | 0D | inf | inf |
| 20 | 2000 | 1D | inf | -3.457 |
| 20 | 2000 | 2D | inf | -8.864 |
| 20 | 2000 | 3D | inf | -9.155 |
| 20 | 2000 | 4D | inf | -14.502 |
| 20 | 2000 | 5D | inf | -14.436 |
| 20 | 2000 | 6D | -4.462 | -25.642 |
| 20 | 2000 | 8D | -24.835 | -29.860 |
| 20 | 2000 | global | inf | inf |
| 50 | 2000 | 0D | inf | inf |
| 50 | 2000 | 1D | -3.753 | -3.753 |
| 50 | 2000 | 2D | inf | -3.842 |
| 50 | 2000 | 3D | inf | -3.986 |
| 50 | 2000 | 4D | inf | inf |
| 50 | 2000 | 5D | inf | inf |
| 50 | 2000 | 6D | inf | -20.434 |
| 50 | 2000 | 8D | inf | -29.960 |
| 50 | 2000 | global | inf | inf |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | inf | inf |
| 100 | 2000 | 2D | -8.927 | -13.727 |
| 100 | 2000 | 3D | -9.220 | -14.162 |
| 100 | 2000 | 4D | -14.604 | -14.604 |
| 100 | 2000 | 5D | -14.538 | -14.538 |
| 100 | 2000 | 6D | -4.660 | -25.192 |
| 100 | 2000 | 8D | inf | -13.564 |
| 100 | 2000 | global | inf | -4.023 |

## 2.2 concentric

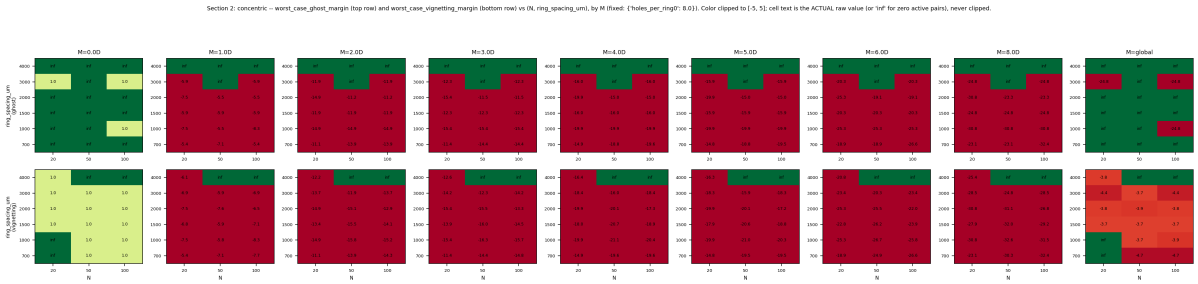



Figure 2.4: concentric: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \text{ring_spacing_um})$, by $M$. holes_per_ring0=8.0 held fixed at its own median tested value; axes: $N \in \{20, 50, 100\}$.

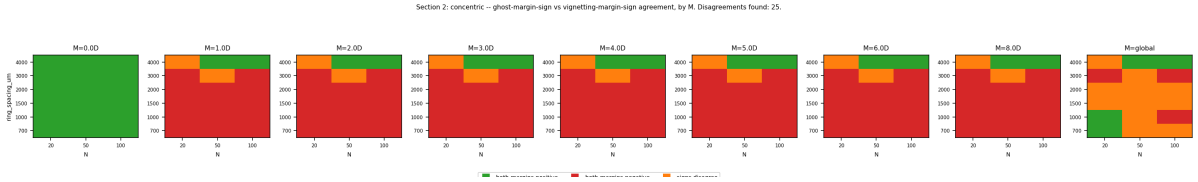



Figure 2.5: concentric: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

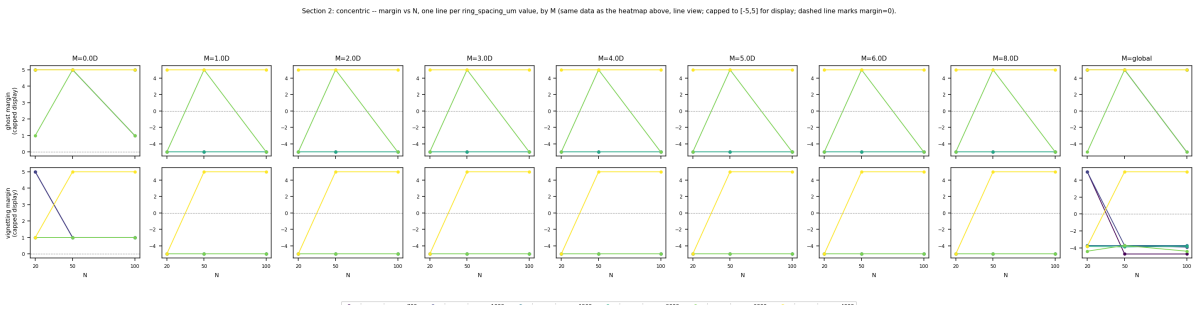



Figure 2.6: concentric: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per ring_spacing_um value) – a complementary graphical view for reading trends across $N$ directly.

### concentric: raw number list

Table 2.2: concentric: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, ring_spacing_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 700 | 0D | inf | inf |
| 20 | 700 | 1D | -5.352 | -5.352 |
| 20 | 700 | 2D | -11.052 | -11.052 |
| 20 | 700 | 3D | -11.408 | -11.408 |
| 20 | 700 | 4D | -14.870 | -14.870 |
| 20 | 700 | 5D | -14.802 | -14.802 |
| 20 | 700 | 6D | -18.904 | -18.904 |
| 20 | 700 | 8D | -23.100 | -23.100 |
| 20 | 700 | global | inf | inf |
| 50 | 700 | 0D | inf | 1.000 |
| 50 | 700 | 1D | -7.096 | -7.096 |
| 50 | 700 | 2D | -13.942 | -13.942 |
| 50 | 700 | 3D | -14.383 | -14.383 |
| 50 | 700 | 4D | -18.837 | -19.615 |
| 50 | 700 | 5D | -18.753 | -19.528 |
| 50 | 700 | 6D | -18.904 | -24.856 |
| 50 | 700 | 8D | -23.100 | -30.307 |
| 50 | 700 | global | inf | -4.733 |
| 100 | 700 | 0D | inf | 1.000 |
| 100 | 700 | 1D | -5.352 | -7.703 |
| 100 | 700 | 2D | -13.942 | -14.326 |
| 100 | 700 | 3D | -14.383 | -14.779 |
| 100 | 700 | 4D | -19.615 | -19.615 |
| 100 | 700 | 5D | -19.528 | -19.528 |
| 100 | 700 | 6D | -26.583 | -26.583 |
| 100 | 700 | 8D | -32.399 | -32.399 |
| 100 | 700 | global | inf | -4.733 |
| 20 | 1000 | 0D | inf | inf |
| 20 | 1000 | 1D | -7.548 | -7.548 |
| 20 | 1000 | 2D | -14.907 | -14.907 |
| 20 | 1000 | 3D | -15.377 | -15.377 |
| 20 | 1000 | 4D | -19.945 | -19.945 |
| 20 | 1000 | 5D | -19.857 | -19.857 |
| 20 | 1000 | 6D | -25.270 | -25.270 |
| 20 | 1000 | 8D | -30.808 | -30.808 |
| 20 | 1000 | global | inf | inf |
| 50 | 1000 | 0D | inf | 1.000 |
| 50 | 1000 | 1D | -5.542 | -5.817 |
| 50 | 1000 | 2D | -14.907 | -15.779 |
| 50 | 1000 | 3D | -15.377 | -16.275 |
| 50 | 1000 | 4D | -19.945 | -21.094 |

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| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1000 | 5D | -19.857 | -21.001 |
| 50 | 1000 | 6D | -25.270 | -26.711 |
| 50 | 1000 | 8D | -30.808 | -32.553 |
| 50 | 1000 | global | inf | -3.729 |
| 100 | 1000 | 0D | 1.000 | 1.000 |
| 100 | 1000 | 1D | -8.252 | -8.252 |
| 100 | 1000 | 2D | -14.907 | -15.244 |
| 100 | 1000 | 3D | -15.377 | -15.725 |
| 100 | 1000 | 4D | -19.945 | -20.390 |
| 100 | 1000 | 5D | -19.857 | -20.299 |
| 100 | 1000 | 6D | -25.270 | -25.827 |
| 100 | 1000 | 8D | -30.808 | -31.483 |
| 100 | 1000 | global | -24.822 | -3.916 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | -5.939 | -6.758 |
| 20 | 1500 | 2D | -11.913 | -13.437 |
| 20 | 1500 | 3D | -12.295 | -13.864 |
| 20 | 1500 | 4D | -16.003 | -18.010 |
| 20 | 1500 | 5D | -15.931 | -17.930 |
| 20 | 1500 | 6D | -20.326 | -22.843 |
| 20 | 1500 | 8D | -24.822 | -27.870 |
| 20 | 1500 | global | inf | -3.729 |
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | -5.939 | -5.939 |
| 50 | 1500 | 2D | -11.913 | -15.500 |
| 50 | 1500 | 3D | -12.295 | -15.988 |
| 50 | 1500 | 4D | -16.003 | -20.726 |
| 50 | 1500 | 5D | -15.931 | -20.634 |
| 50 | 1500 | 6D | -20.326 | -26.249 |
| 50 | 1500 | 8D | -24.822 | -31.994 |
| 50 | 1500 | global | inf | -3.729 |
| 100 | 1500 | 0D | inf | 1.000 |
| 100 | 1500 | 1D | -5.939 | -7.113 |
| 100 | 1500 | 2D | -11.913 | -14.098 |
| 100 | 1500 | 3D | -12.295 | -14.545 |
| 100 | 1500 | 4D | -16.003 | -18.881 |
| 100 | 1500 | 5D | -15.931 | -18.796 |
| 100 | 1500 | 6D | -20.326 | -23.934 |
| 100 | 1500 | 8D | -24.822 | -29.191 |
| 100 | 1500 | global | inf | -3.729 |
| 20 | 2000 | 0D | inf | 1.000 |
| 20 | 2000 | 1D | -7.548 | -7.548 |
| 20 | 2000 | 2D | -14.907 | -14.907 |
| 20 | 2000 | 3D | -15.377 | -15.377 |
| 20 | 2000 | 4D | -19.945 | -19.945 |
| 20 | 2000 | 5D | -19.857 | -19.857 |
| 20 | 2000 | 6D | -25.270 | -25.270 |
| 20 | 2000 | 8D | -30.808 | -30.808 |
| 20 | 2000 | global | inf | -3.825 |

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| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 2000 | 0D | inf | 1.000 |
| 50 | 2000 | 1D | -5.542 | -7.630 |
| 50 | 2000 | 2D | -11.175 | -15.059 |
| 50 | 2000 | 3D | -11.534 | -15.534 |
| 50 | 2000 | 4D | -15.031 | -20.145 |
| 50 | 2000 | 5D | -14.963 | -20.056 |
| 50 | 2000 | 6D | -19.106 | -25.521 |
| 50 | 2000 | 8D | -23.345 | -31.112 |
| 50 | 2000 | global | inf | -3.895 |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | -5.542 | -6.470 |
| 100 | 2000 | 2D | -11.175 | -12.902 |
| 100 | 2000 | 3D | -11.534 | -13.313 |
| 100 | 2000 | 4D | -15.031 | -17.305 |
| 100 | 2000 | 5D | -14.963 | -17.227 |
| 100 | 2000 | 6D | -19.106 | -21.958 |
| 100 | 2000 | 8D | -23.345 | -26.799 |
| 100 | 2000 | global | inf | -3.825 |
| 20 | 3000 | 0D | 1.000 | 1.000 |
| 20 | 3000 | 1D | -5.939 | -6.926 |
| 20 | 3000 | 2D | -11.913 | -13.748 |
| 20 | 3000 | 3D | -12.295 | -14.185 |
| 20 | 3000 | 4D | -16.003 | -18.420 |
| 20 | 3000 | 5D | -15.931 | -18.338 |
| 20 | 3000 | 6D | -20.326 | -23.357 |
| 20 | 3000 | 8D | -24.822 | -28.492 |
| 20 | 3000 | global | -24.822 | -4.401 |
| 50 | 3000 | 0D | inf | 1.000 |
| 50 | 3000 | 1D | inf | -5.939 |
| 50 | 3000 | 2D | inf | -11.913 |
| 50 | 3000 | 3D | inf | -12.295 |
| 50 | 3000 | 4D | inf | -16.003 |
| 50 | 3000 | 5D | inf | -15.931 |
| 50 | 3000 | 6D | inf | -20.326 |
| 50 | 3000 | 8D | inf | -24.822 |
| 50 | 3000 | global | inf | -3.729 |
| 100 | 3000 | 0D | 1.000 | 1.000 |
| 100 | 3000 | 1D | -5.939 | -6.926 |
| 100 | 3000 | 2D | -11.913 | -13.748 |
| 100 | 3000 | 3D | -12.295 | -14.185 |
| 100 | 3000 | 4D | -16.003 | -18.420 |
| 100 | 3000 | 5D | -15.931 | -18.338 |
| 100 | 3000 | 6D | -20.326 | -23.357 |
| 100 | 3000 | 8D | -24.822 | -28.492 |
| 100 | 3000 | global | -24.822 | -4.401 |
| 20 | 4000 | 0D | inf | 1.000 |
| 20 | 4000 | 1D | inf | -6.081 |
| 20 | 4000 | 2D | inf | -12.178 |
| 20 | 4000 | 3D | inf | -12.567 |

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| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 4000 | 4D | inf | -16.352 |
| 20 | 4000 | 5D | inf | -16.278 |
| 20 | 4000 | 6D | inf | -20.763 |
| 20 | 4000 | 8D | inf | -25.351 |
| 20 | 4000 | global | inf | -3.825 |
| 50 | 4000 | 0D | inf | inf |
| 50 | 4000 | 1D | inf | inf |
| 50 | 4000 | 2D | inf | inf |
| 50 | 4000 | 3D | inf | inf |
| 50 | 4000 | 4D | inf | inf |
| 50 | 4000 | 5D | inf | inf |
| 50 | 4000 | 6D | inf | inf |
| 50 | 4000 | 8D | inf | inf |
| 50 | 4000 | global | inf | inf |
| 100 | 4000 | 0D | inf | inf |
| 100 | 4000 | 1D | inf | inf |
| 100 | 4000 | 2D | inf | inf |
| 100 | 4000 | 3D | inf | inf |
| 100 | 4000 | 4D | inf | inf |
| 100 | 4000 | 5D | inf | inf |
| 100 | 4000 | 6D | inf | inf |
| 100 | 4000 | 8D | inf | inf |
| 100 | 4000 | global | inf | inf |

## 2.3 fibonacci_spiral

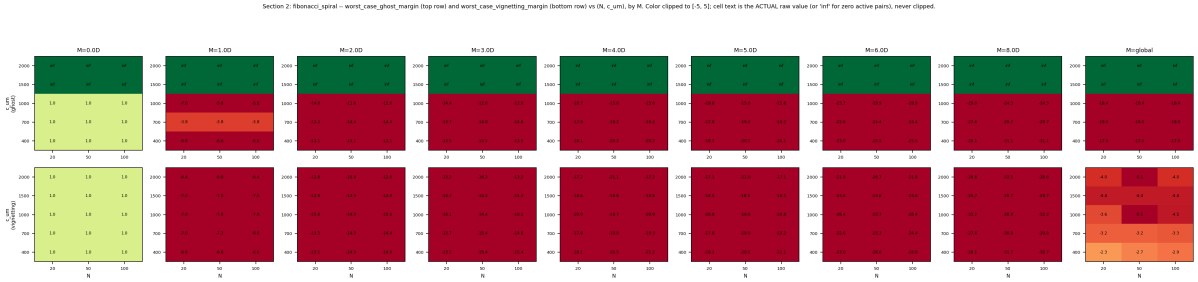



Figure 2.7: fibonacci_spiral: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \mathrm{c_um})$, by $M$. axes: $N \in \{20, 50, 100\}$.

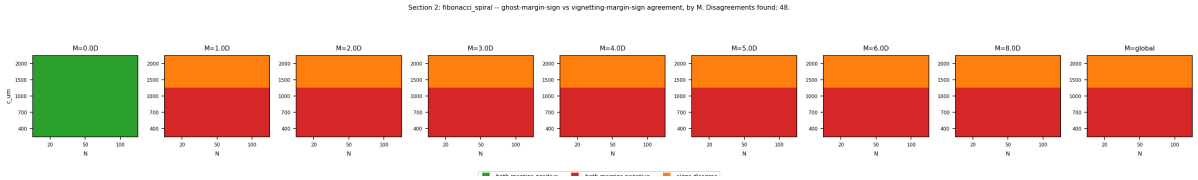



Figure 2.8: fibonacci_spiral: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

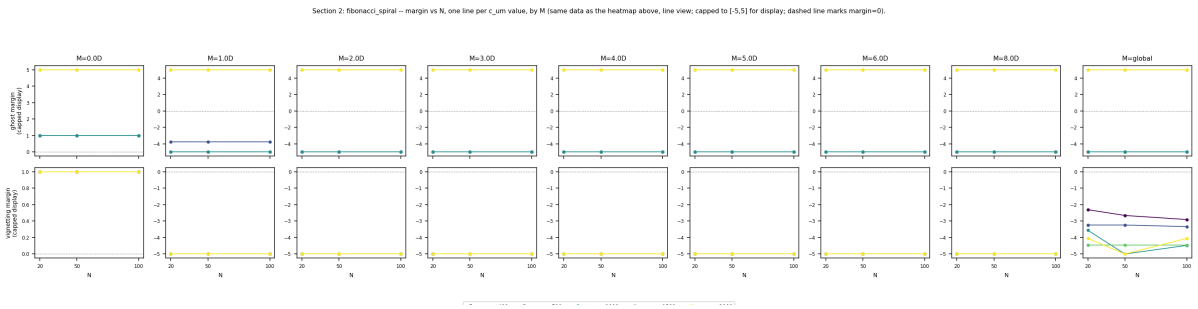



Figure 2.9: fibonacci_spiral: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per c_um value) – a complementary graphical view for reading trends across $N$ directly.

### fibonacci_spiral: raw number list

Table 2.3: fibonacci_spiral: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, c_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 400 | 0D | 1.000 | 1.000 |
| 20 | 400 | 1D | -6.587 | -6.587 |
| 20 | 400 | 2D | -13.119 | -13.119 |
| 20 | 400 | 3D | -13.536 | -13.536 |
| 20 | 400 | 4D | -18.134 | -18.134 |
| 20 | 400 | 5D | -18.053 | -18.053 |
| 20 | 400 | 6D | -22.998 | -22.998 |
| 20 | 400 | 8D | -28.057 | -28.057 |
| 20 | 400 | global | -17.047 | -2.305 |
| 50 | 400 | 0D | 1.000 | 1.000 |
| 50 | 400 | 1D | -6.587 | -6.587 |
| 50 | 400 | 2D | -13.119 | -14.928 |
| 50 | 400 | 3D | -13.536 | -15.399 |
| 50 | 400 | 4D | -20.164 | -20.542 |
| 50 | 400 | 5D | -20.074 | -20.451 |
| 50 | 400 | 6D | -25.544 | -26.018 |
| 50 | 400 | 8D | -31.140 | -31.715 |
| 50 | 400 | global | -17.047 | -2.657 |
| 100 | 400 | 0D | 1.000 | 1.000 |
| 100 | 400 | 1D | -6.587 | -6.587 |
| 100 | 400 | 2D | -13.119 | -14.929 |
| 100 | 400 | 3D | -13.536 | -15.400 |
| 100 | 400 | 4D | -20.164 | -21.185 |
| 100 | 400 | 5D | -20.074 | -21.091 |
| 100 | 400 | 6D | -25.544 | -26.825 |
| 100 | 400 | 8D | -31.140 | -32.691 |
| 100 | 400 | global | -17.047 | -2.904 |
| 20 | 700 | 0D | 1.000 | 1.000 |
| 20 | 700 | 1D | -3.752 | -7.016 |
| 20 | 700 | 2D | -13.310 | -13.310 |
| 20 | 700 | 3D | -13.733 | -13.733 |
| 20 | 700 | 4D | -17.843 | -17.843 |
| 20 | 700 | 5D | -17.763 | -17.763 |
| 20 | 700 | 6D | -22.633 | -22.633 |
| 20 | 700 | 8D | -27.616 | -27.616 |
| 20 | 700 | global | -18.952 | -3.238 |
| 50 | 700 | 0D | 1.000 | 1.000 |
| 50 | 700 | 1D | -3.752 | -7.268 |
| 50 | 700 | 2D | -14.374 | -14.907 |
| 50 | 700 | 3D | -14.829 | -15.377 |
| 50 | 700 | 4D | -19.244 | -19.946 |

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| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 700 | 5D | -19.158 | -19.857 |
| 50 | 700 | 6D | -24.390 | -25.270 |
| 50 | 700 | 8D | -29.743 | -30.809 |
| 50 | 700 | global | -18.952 | -3.238 |
| 100 | 700 | 0D | 1.000 | 1.000 |
| 100 | 700 | 1D | -3.752 | -7.974 |
| 100 | 700 | 2D | -14.374 | -14.386 |
| 100 | 700 | 3D | -14.829 | -14.841 |
| 100 | 700 | 4D | -19.244 | -19.259 |
| 100 | 700 | 5D | -19.158 | -19.174 |
| 100 | 700 | 6D | -24.390 | -24.410 |
| 100 | 700 | 8D | -29.743 | -29.767 |
| 100 | 700 | global | -18.952 | -3.333 |
| 20 | 1000 | 0D | 1.000 | 1.000 |
| 20 | 1000 | 1D | -7.049 | -7.932 |
| 20 | 1000 | 2D | -13.978 | -15.621 |
| 20 | 1000 | 3D | -14.421 | -16.112 |
| 20 | 1000 | 4D | -18.722 | -20.886 |
| 20 | 1000 | 5D | -18.639 | -20.793 |
| 20 | 1000 | 6D | -23.736 | -26.449 |
| 20 | 1000 | 8D | -28.951 | -32.237 |
| 20 | 1000 | global | -18.410 | -3.556 |
| 50 | 1000 | 0D | 1.000 | 1.000 |
| 50 | 1000 | 1D | -5.788 | -7.038 |
| 50 | 1000 | 2D | -11.632 | -13.958 |
| 50 | 1000 | 3D | -12.005 | -14.401 |
| 50 | 1000 | 4D | -15.633 | -18.696 |
| 50 | 1000 | 5D | -15.563 | -18.613 |
| 50 | 1000 | 6D | -19.861 | -23.703 |
| 50 | 1000 | 8D | -24.259 | -28.912 |
| 50 | 1000 | global | -18.410 | -5.086 |
| 100 | 1000 | 0D | 1.000 | 1.000 |
| 100 | 1000 | 1D | -5.788 | -7.932 |
| 100 | 1000 | 2D | -11.632 | -15.621 |
| 100 | 1000 | 3D | -12.005 | -16.112 |
| 100 | 1000 | 4D | -15.633 | -20.886 |
| 100 | 1000 | 5D | -15.563 | -20.793 |
| 100 | 1000 | 6D | -19.861 | -26.449 |
| 100 | 1000 | 8D | -24.259 | -32.237 |
| 100 | 1000 | global | -18.410 | -4.478 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | inf | -6.991 |
| 20 | 1500 | 2D | inf | -13.869 |
| 20 | 1500 | 3D | inf | -14.309 |
| 20 | 1500 | 4D | inf | -18.579 |
| 20 | 1500 | 5D | inf | -18.496 |
| 20 | 1500 | 6D | inf | -23.556 |
| 20 | 1500 | 8D | inf | -28.734 |
| 20 | 1500 | global | inf | -4.445 |

*continued on next page*

| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | inf | -6.991 |
| 50 | 1500 | 2D | inf | -13.869 |
| 50 | 1500 | 3D | inf | -14.309 |
| 50 | 1500 | 4D | inf | -18.579 |
| 50 | 1500 | 5D | inf | -18.496 |
| 50 | 1500 | 6D | inf | -23.556 |
| 50 | 1500 | 8D | inf | -28.734 |
| 50 | 1500 | global | inf | -4.445 |
| 100 | 1500 | 0D | inf | 1.000 |
| 100 | 1500 | 1D | inf | -6.991 |
| 100 | 1500 | 2D | inf | -13.869 |
| 100 | 1500 | 3D | inf | -14.309 |
| 100 | 1500 | 4D | inf | -18.579 |
| 100 | 1500 | 5D | inf | -18.496 |
| 100 | 1500 | 6D | inf | -23.556 |
| 100 | 1500 | 8D | inf | -28.734 |
| 100 | 1500 | global | inf | -4.445 |
| 20 | 2000 | 0D | inf | 1.000 |
| 20 | 2000 | 1D | inf | -6.411 |
| 20 | 2000 | 2D | inf | -12.791 |
| 20 | 2000 | 3D | inf | -13.198 |
| 20 | 2000 | 4D | inf | -17.159 |
| 20 | 2000 | 5D | inf | -17.082 |
| 20 | 2000 | 6D | inf | -21.775 |
| 20 | 2000 | 8D | inf | -26.577 |
| 20 | 2000 | global | inf | -4.050 |
| 50 | 2000 | 0D | inf | 1.000 |
| 50 | 2000 | 1D | inf | -8.007 |
| 50 | 2000 | 2D | inf | -15.761 |
| 50 | 2000 | 3D | inf | -16.256 |
| 50 | 2000 | 4D | inf | -21.070 |
| 50 | 2000 | 5D | inf | -20.976 |
| 50 | 2000 | 6D | inf | -26.680 |
| 50 | 2000 | 8D | inf | -32.516 |
| 50 | 2000 | global | inf | -5.138 |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | inf | -6.411 |
| 100 | 2000 | 2D | inf | -12.791 |
| 100 | 2000 | 3D | inf | -13.198 |
| 100 | 2000 | 4D | inf | -17.159 |
| 100 | 2000 | 5D | inf | -17.082 |
| 100 | 2000 | 6D | inf | -21.775 |
| 100 | 2000 | 8D | inf | -26.577 |
| 100 | 2000 | global | inf | -4.050 |

## 2.4 golden_angle

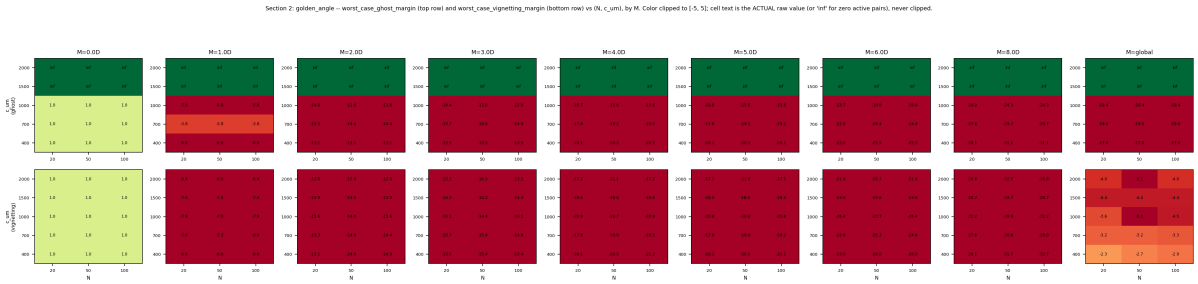



Figure 2.10: golden_angle: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \mathrm{c_um})$, by $M$. axes: $N \in \{20, 50, 100\}$.

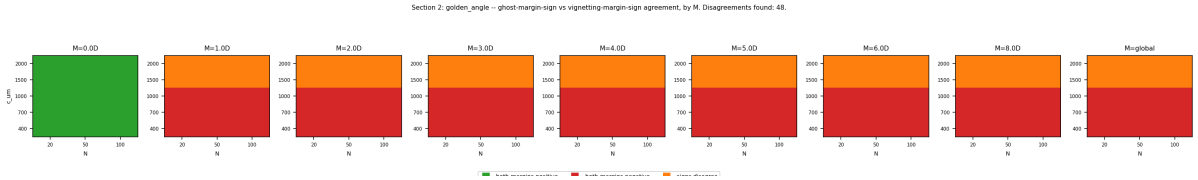



Figure 2.11: golden_angle: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

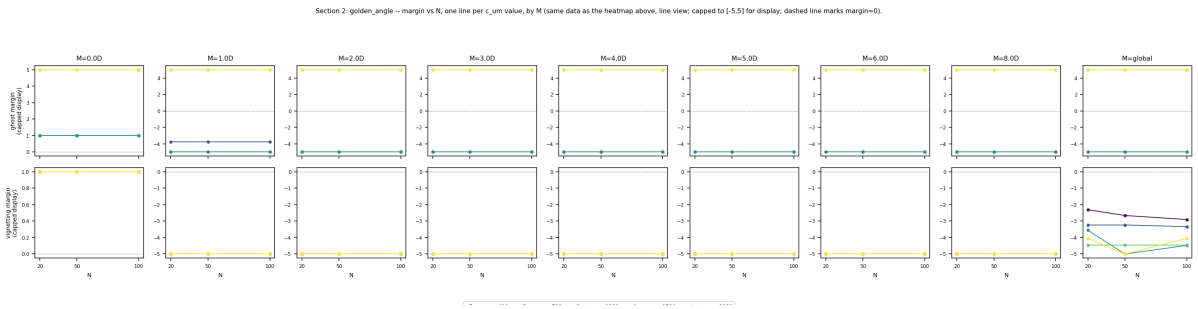



Figure 2.12: golden_angle: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per c_um value) – a complementary graphical view for reading trends across $N$ directly.

### golden_angle: raw number list

Table 2.4: golden_angle: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, c_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 400 | 0D | 1.000 | 1.000 |
| 20 | 400 | 1D | -6.587 | -6.587 |
| 20 | 400 | 2D | -13.119 | -13.119 |
| 20 | 400 | 3D | -13.536 | -13.536 |
| 20 | 400 | 4D | -18.134 | -18.134 |
| 20 | 400 | 5D | -18.053 | -18.053 |
| 20 | 400 | 6D | -22.998 | -22.998 |
| 20 | 400 | 8D | -28.057 | -28.057 |
| 20 | 400 | global | -17.047 | -2.305 |
| 50 | 400 | 0D | 1.000 | 1.000 |
| 50 | 400 | 1D | -6.587 | -6.587 |
| 50 | 400 | 2D | -13.119 | -14.928 |
| 50 | 400 | 3D | -13.536 | -15.399 |
| 50 | 400 | 4D | -20.164 | -20.542 |
| 50 | 400 | 5D | -20.074 | -20.451 |
| 50 | 400 | 6D | -25.544 | -26.018 |
| 50 | 400 | 8D | -31.140 | -31.715 |
| 50 | 400 | global | -17.047 | -2.657 |
| 100 | 400 | 0D | 1.000 | 1.000 |
| 100 | 400 | 1D | -6.587 | -6.587 |
| 100 | 400 | 2D | -13.119 | -14.929 |
| 100 | 400 | 3D | -13.536 | -15.400 |
| 100 | 400 | 4D | -20.164 | -21.185 |
| 100 | 400 | 5D | -20.074 | -21.091 |
| 100 | 400 | 6D | -25.544 | -26.825 |
| 100 | 400 | 8D | -31.140 | -32.691 |
| 100 | 400 | global | -17.047 | -2.904 |
| 20 | 700 | 0D | 1.000 | 1.000 |
| 20 | 700 | 1D | -3.752 | -7.016 |
| 20 | 700 | 2D | -13.310 | -13.310 |
| 20 | 700 | 3D | -13.733 | -13.733 |
| 20 | 700 | 4D | -17.843 | -17.843 |
| 20 | 700 | 5D | -17.763 | -17.763 |
| 20 | 700 | 6D | -22.633 | -22.633 |
| 20 | 700 | 8D | -27.616 | -27.616 |
| 20 | 700 | global | -18.952 | -3.238 |
| 50 | 700 | 0D | 1.000 | 1.000 |
| 50 | 700 | 1D | -3.752 | -7.268 |
| 50 | 700 | 2D | -14.374 | -14.907 |
| 50 | 700 | 3D | -14.829 | -15.377 |
| 50 | 700 | 4D | -19.244 | -19.946 |

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| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 700 | 5D | -19.158 | -19.857 |
| 50 | 700 | 6D | -24.390 | -25.270 |
| 50 | 700 | 8D | -29.743 | -30.809 |
| 50 | 700 | global | -18.952 | -3.238 |
| 100 | 700 | 0D | 1.000 | 1.000 |
| 100 | 700 | 1D | -3.752 | -7.974 |
| 100 | 700 | 2D | -14.374 | -14.386 |
| 100 | 700 | 3D | -14.829 | -14.841 |
| 100 | 700 | 4D | -19.244 | -19.259 |
| 100 | 700 | 5D | -19.158 | -19.174 |
| 100 | 700 | 6D | -24.390 | -24.410 |
| 100 | 700 | 8D | -29.743 | -29.767 |
| 100 | 700 | global | -18.952 | -3.333 |
| 20 | 1000 | 0D | 1.000 | 1.000 |
| 20 | 1000 | 1D | -7.049 | -7.932 |
| 20 | 1000 | 2D | -13.978 | -15.621 |
| 20 | 1000 | 3D | -14.421 | -16.112 |
| 20 | 1000 | 4D | -18.722 | -20.886 |
| 20 | 1000 | 5D | -18.639 | -20.793 |
| 20 | 1000 | 6D | -23.736 | -26.449 |
| 20 | 1000 | 8D | -28.951 | -32.237 |
| 20 | 1000 | global | -18.410 | -3.556 |
| 50 | 1000 | 0D | 1.000 | 1.000 |
| 50 | 1000 | 1D | -5.788 | -7.038 |
| 50 | 1000 | 2D | -11.632 | -13.958 |
| 50 | 1000 | 3D | -12.005 | -14.401 |
| 50 | 1000 | 4D | -15.633 | -18.696 |
| 50 | 1000 | 5D | -15.563 | -18.613 |
| 50 | 1000 | 6D | -19.861 | -23.703 |
| 50 | 1000 | 8D | -24.259 | -28.912 |
| 50 | 1000 | global | -18.410 | -5.086 |
| 100 | 1000 | 0D | 1.000 | 1.000 |
| 100 | 1000 | 1D | -5.788 | -7.932 |
| 100 | 1000 | 2D | -11.632 | -15.621 |
| 100 | 1000 | 3D | -12.005 | -16.112 |
| 100 | 1000 | 4D | -15.633 | -20.886 |
| 100 | 1000 | 5D | -15.563 | -20.793 |
| 100 | 1000 | 6D | -19.861 | -26.449 |
| 100 | 1000 | 8D | -24.259 | -32.237 |
| 100 | 1000 | global | -18.410 | -4.478 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | inf | -6.991 |
| 20 | 1500 | 2D | inf | -13.869 |
| 20 | 1500 | 3D | inf | -14.309 |
| 20 | 1500 | 4D | inf | -18.579 |
| 20 | 1500 | 5D | inf | -18.496 |
| 20 | 1500 | 6D | inf | -23.556 |
| 20 | 1500 | 8D | inf | -28.734 |
| 20 | 1500 | global | inf | -4.445 |

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| N | c_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | inf | -6.991 |
| 50 | 1500 | 2D | inf | -13.869 |
| 50 | 1500 | 3D | inf | -14.309 |
| 50 | 1500 | 4D | inf | -18.579 |
| 50 | 1500 | 5D | inf | -18.496 |
| 50 | 1500 | 6D | inf | -23.556 |
| 50 | 1500 | 8D | inf | -28.734 |
| 50 | 1500 | global | inf | -4.445 |
| 100 | 1500 | 0D | inf | 1.000 |
| 100 | 1500 | 1D | inf | -6.991 |
| 100 | 1500 | 2D | inf | -13.869 |
| 100 | 1500 | 3D | inf | -14.309 |
| 100 | 1500 | 4D | inf | -18.579 |
| 100 | 1500 | 5D | inf | -18.496 |
| 100 | 1500 | 6D | inf | -23.556 |
| 100 | 1500 | 8D | inf | -28.734 |
| 100 | 1500 | global | inf | -4.445 |
| 20 | 2000 | 0D | inf | 1.000 |
| 20 | 2000 | 1D | inf | -6.411 |
| 20 | 2000 | 2D | inf | -12.791 |
| 20 | 2000 | 3D | inf | -13.198 |
| 20 | 2000 | 4D | inf | -17.159 |
| 20 | 2000 | 5D | inf | -17.082 |
| 20 | 2000 | 6D | inf | -21.775 |
| 20 | 2000 | 8D | inf | -26.577 |
| 20 | 2000 | global | inf | -4.050 |
| 50 | 2000 | 0D | inf | 1.000 |
| 50 | 2000 | 1D | inf | -8.007 |
| 50 | 2000 | 2D | inf | -15.761 |
| 50 | 2000 | 3D | inf | -16.256 |
| 50 | 2000 | 4D | inf | -21.070 |
| 50 | 2000 | 5D | inf | -20.976 |
| 50 | 2000 | 6D | inf | -26.680 |
| 50 | 2000 | 8D | inf | -32.516 |
| 50 | 2000 | global | inf | -5.138 |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | inf | -6.411 |
| 100 | 2000 | 2D | inf | -12.791 |
| 100 | 2000 | 3D | inf | -13.198 |
| 100 | 2000 | 4D | inf | -17.159 |
| 100 | 2000 | 5D | inf | -17.082 |
| 100 | 2000 | 6D | inf | -21.775 |
| 100 | 2000 | 8D | inf | -26.577 |
| 100 | 2000 | global | inf | -4.050 |

## 2.5 hexagonal

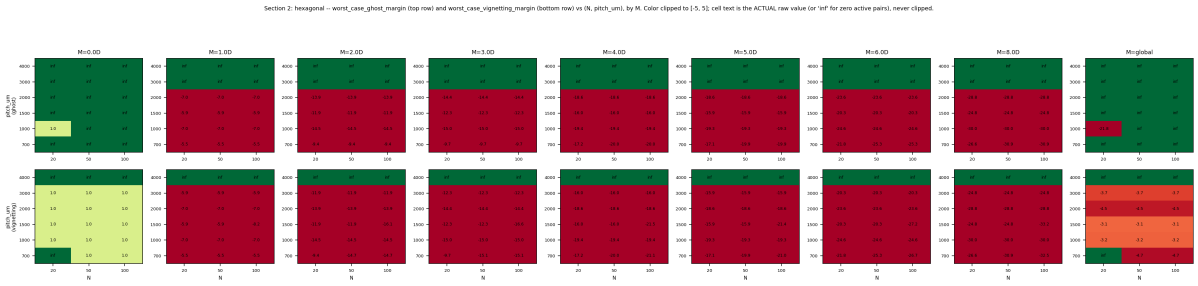


Figure 2.13: hexagonal: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. ($N$, pitch_um), by $M$. axes: $N \in \{20, 50, 100\}$.

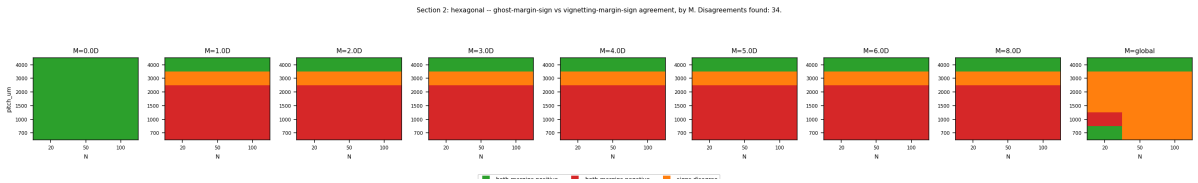


Figure 2.14: hexagonal: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

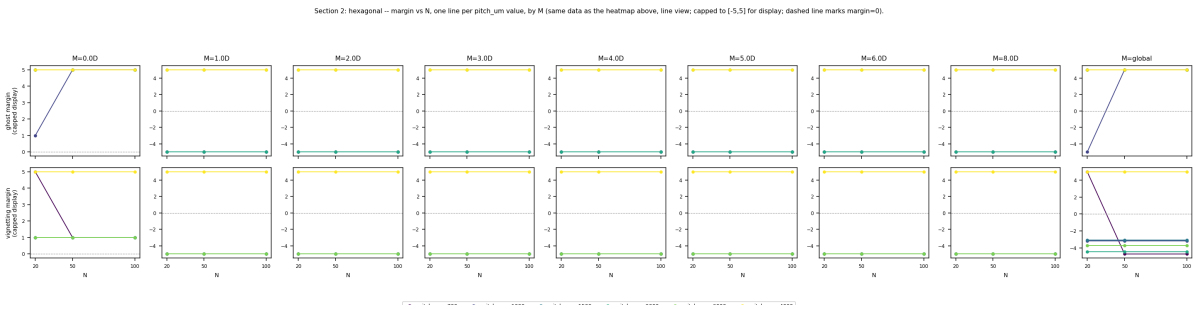



Figure 2.15: hexagonal: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per pitch_um value) – a complementary graphical view for reading trends across $N$ directly.

### hexagonal: raw number list

Table 2.5: hexagonal: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, pitch_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 700 | 0D | inf | inf |
| 20 | 700 | 1D | -5.477 | -5.477 |
| 20 | 700 | 2D | -9.437 | -9.437 |
| 20 | 700 | 3D | -9.746 | -9.746 |
| 20 | 700 | 4D | -17.181 | -17.181 |
| 20 | 700 | 5D | -17.104 | -17.104 |
| 20 | 700 | 6D | -21.803 | -21.803 |
| 20 | 700 | 8D | -26.610 | -26.610 |
| 20 | 700 | global | inf | inf |
| 50 | 700 | 0D | inf | 1.000 |
| 50 | 700 | 1D | -5.477 | -5.477 |
| 50 | 700 | 2D | -9.437 | -14.656 |
| 50 | 700 | 3D | -9.746 | -15.119 |
| 50 | 700 | 4D | -19.994 | -19.994 |
| 50 | 700 | 5D | -19.905 | -19.905 |
| 50 | 700 | 6D | -25.330 | -25.330 |
| 50 | 700 | 8D | -30.882 | -30.882 |
| 50 | 700 | global | inf | -4.733 |
| 100 | 700 | 0D | inf | 1.000 |
| 100 | 700 | 1D | -5.477 | -5.477 |
| 100 | 700 | 2D | -9.437 | -14.656 |
| 100 | 700 | 3D | -9.746 | -15.119 |
| 100 | 700 | 4D | -19.994 | -21.090 |
| 100 | 700 | 5D | -19.905 | -20.996 |
| 100 | 700 | 6D | -25.330 | -26.705 |
| 100 | 700 | 8D | -30.882 | -32.546 |
| 100 | 700 | global | inf | -4.733 |
| 20 | 1000 | 0D | 1.000 | 1.000 |
| 20 | 1000 | 1D | -7.013 | -7.013 |
| 20 | 1000 | 2D | -14.520 | -14.520 |
| 20 | 1000 | 3D | -14.978 | -14.978 |
| 20 | 1000 | 4D | -19.435 | -19.435 |
| 20 | 1000 | 5D | -19.349 | -19.349 |
| 20 | 1000 | 6D | -24.630 | -24.630 |
| 20 | 1000 | 8D | -30.034 | -30.034 |
| 20 | 1000 | global | -21.773 | -3.170 |
| 50 | 1000 | 0D | inf | 1.000 |
| 50 | 1000 | 1D | -7.013 | -7.013 |
| 50 | 1000 | 2D | -14.520 | -14.520 |
| 50 | 1000 | 3D | -14.978 | -14.978 |
| 50 | 1000 | 4D | -19.435 | -19.435 |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1000 | 5D | -19.349 | -19.349 |
| 50 | 1000 | 6D | -24.630 | -24.630 |
| 50 | 1000 | 8D | -30.034 | -30.034 |
| 50 | 1000 | global | inf | -3.170 |
| 100 | 1000 | 0D | inf | 1.000 |
| 100 | 1000 | 1D | -7.013 | -7.013 |
| 100 | 1000 | 2D | -14.520 | -14.520 |
| 100 | 1000 | 3D | -14.978 | -14.978 |
| 100 | 1000 | 4D | -19.435 | -19.435 |
| 100 | 1000 | 5D | -19.349 | -19.349 |
| 100 | 1000 | 6D | -24.630 | -24.630 |
| 100 | 1000 | 8D | -30.034 | -30.034 |
| 100 | 1000 | global | inf | -3.170 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | -5.939 | -5.939 |
| 20 | 1500 | 2D | -11.913 | -11.913 |
| 20 | 1500 | 3D | -12.295 | -12.295 |
| 20 | 1500 | 4D | -16.003 | -16.003 |
| 20 | 1500 | 5D | -15.931 | -15.931 |
| 20 | 1500 | 6D | -20.326 | -20.326 |
| 20 | 1500 | 8D | -24.822 | -24.822 |
| 20 | 1500 | global | inf | -3.095 |
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | -5.939 | -5.939 |
| 50 | 1500 | 2D | -11.913 | -11.913 |
| 50 | 1500 | 3D | -12.295 | -12.295 |
| 50 | 1500 | 4D | -16.003 | -16.003 |
| 50 | 1500 | 5D | -15.931 | -15.931 |
| 50 | 1500 | 6D | -20.326 | -20.326 |
| 50 | 1500 | 8D | -24.822 | -24.822 |
| 50 | 1500 | global | inf | -3.095 |
| 100 | 1500 | 0D | inf | 1.000 |
| 100 | 1500 | 1D | -5.939 | -8.180 |
| 100 | 1500 | 2D | -11.913 | -16.082 |
| 100 | 1500 | 3D | -12.295 | -16.587 |
| 100 | 1500 | 4D | -16.003 | -21.493 |
| 100 | 1500 | 5D | -15.931 | -21.398 |
| 100 | 1500 | 6D | -20.326 | -27.211 |
| 100 | 1500 | 8D | -24.822 | -33.159 |
| 100 | 1500 | global | inf | -3.095 |
| 20 | 2000 | 0D | inf | 1.000 |
| 20 | 2000 | 1D | -7.013 | -7.013 |
| 20 | 2000 | 2D | -13.911 | -13.911 |
| 20 | 2000 | 3D | -14.352 | -14.352 |
| 20 | 2000 | 4D | -18.634 | -18.634 |
| 20 | 2000 | 5D | -18.550 | -18.550 |
| 20 | 2000 | 6D | -23.625 | -23.625 |
| 20 | 2000 | 8D | -28.816 | -28.816 |
| 20 | 2000 | global | inf | -4.460 |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 2000 | 0D | inf | 1.000 |
| 50 | 2000 | 1D | -7.013 | -7.013 |
| 50 | 2000 | 2D | -13.911 | -13.911 |
| 50 | 2000 | 3D | -14.352 | -14.352 |
| 50 | 2000 | 4D | -18.634 | -18.634 |
| 50 | 2000 | 5D | -18.550 | -18.550 |
| 50 | 2000 | 6D | -23.625 | -23.625 |
| 50 | 2000 | 8D | -28.816 | -28.816 |
| 50 | 2000 | global | inf | -4.460 |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | -7.013 | -7.013 |
| 100 | 2000 | 2D | -13.911 | -13.911 |
| 100 | 2000 | 3D | -14.352 | -14.352 |
| 100 | 2000 | 4D | -18.634 | -18.634 |
| 100 | 2000 | 5D | -18.550 | -18.550 |
| 100 | 2000 | 6D | -23.625 | -23.625 |
| 100 | 2000 | 8D | -28.816 | -28.816 |
| 100 | 2000 | global | inf | -4.460 |
| 20 | 3000 | 0D | inf | 1.000 |
| 20 | 3000 | 1D | inf | -5.939 |
| 20 | 3000 | 2D | inf | -11.913 |
| 20 | 3000 | 3D | inf | -12.295 |
| 20 | 3000 | 4D | inf | -16.003 |
| 20 | 3000 | 5D | inf | -15.931 |
| 20 | 3000 | 6D | inf | -20.326 |
| 20 | 3000 | 8D | inf | -24.822 |
| 20 | 3000 | global | inf | -3.729 |
| 50 | 3000 | 0D | inf | 1.000 |
| 50 | 3000 | 1D | inf | -5.939 |
| 50 | 3000 | 2D | inf | -11.913 |
| 50 | 3000 | 3D | inf | -12.295 |
| 50 | 3000 | 4D | inf | -16.003 |
| 50 | 3000 | 5D | inf | -15.931 |
| 50 | 3000 | 6D | inf | -20.326 |
| 50 | 3000 | 8D | inf | -24.822 |
| 50 | 3000 | global | inf | -3.729 |
| 100 | 3000 | 0D | inf | 1.000 |
| 100 | 3000 | 1D | inf | -5.939 |
| 100 | 3000 | 2D | inf | -11.913 |
| 100 | 3000 | 3D | inf | -12.295 |
| 100 | 3000 | 4D | inf | -16.003 |
| 100 | 3000 | 5D | inf | -15.931 |
| 100 | 3000 | 6D | inf | -20.326 |
| 100 | 3000 | 8D | inf | -24.822 |
| 100 | 3000 | global | inf | -3.729 |
| 20 | 4000 | 0D | inf | inf |
| 20 | 4000 | 1D | inf | inf |
| 20 | 4000 | 2D | inf | inf |
| 20 | 4000 | 3D | inf | inf |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 4000 | 4D | inf | inf |
| 20 | 4000 | 5D | inf | inf |
| 20 | 4000 | 6D | inf | inf |
| 20 | 4000 | 8D | inf | inf |
| 20 | 4000 | global | inf | inf |
| 50 | 4000 | 0D | inf | inf |
| 50 | 4000 | 1D | inf | inf |
| 50 | 4000 | 2D | inf | inf |
| 50 | 4000 | 3D | inf | inf |
| 50 | 4000 | 4D | inf | inf |
| 50 | 4000 | 5D | inf | inf |
| 50 | 4000 | 6D | inf | inf |
| 50 | 4000 | 8D | inf | inf |
| 50 | 4000 | global | inf | inf |
| 100 | 4000 | 0D | inf | inf |
| 100 | 4000 | 1D | inf | inf |
| 100 | 4000 | 2D | inf | inf |
| 100 | 4000 | 3D | inf | inf |
| 100 | 4000 | 4D | inf | inf |
| 100 | 4000 | 5D | inf | inf |
| 100 | 4000 | 6D | inf | inf |
| 100 | 4000 | 8D | inf | inf |
| 100 | 4000 | global | inf | inf |

## 2.6 jittered_lattice

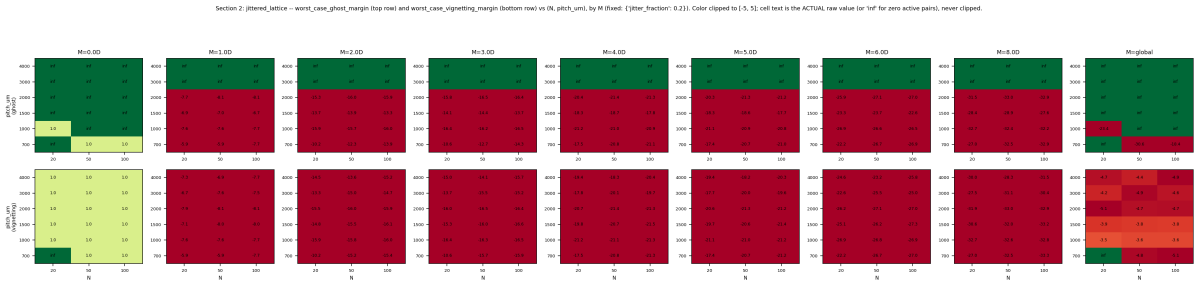



Figure 2.16: jittered_lattice: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \text{pitch_um})$, by $M$. jitter_fraction=0.2 held fixed at its own median tested value; axes: $N \in \{20, 50, 100\}$.

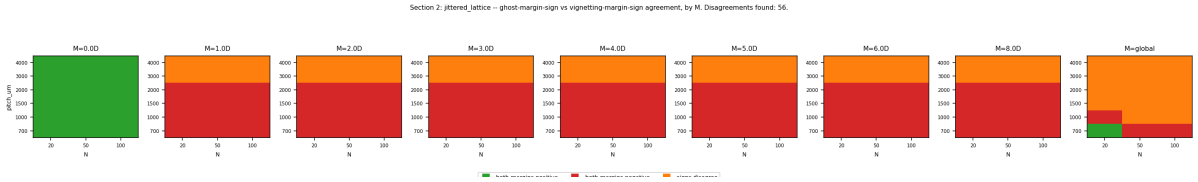



Figure 2.17: jittered_lattice: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

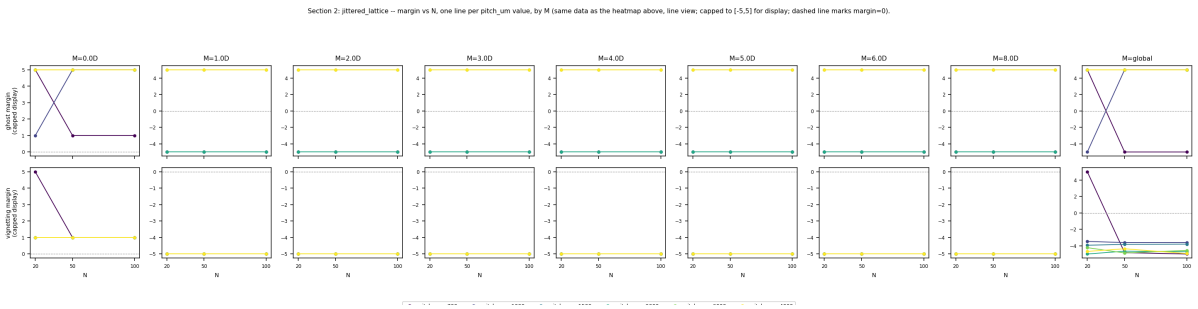



Figure 2.18: jittered_lattice: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per pitch_um value) – a complementary graphical view for reading trends across $N$ directly.

### jittered_lattice: raw number list

Table 2.6: jittered_lattice: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, pitch_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 700 | 0D | inf | inf |
| 20 | 700 | 1D | -5.944 | -5.944 |
| 20 | 700 | 2D | -10.220 | -10.220 |
| 20 | 700 | 3D | -10.551 | -10.551 |
| 20 | 700 | 4D | -17.460 | -17.460 |
| 20 | 700 | 5D | -17.382 | -17.382 |
| 20 | 700 | 6D | -22.153 | -22.153 |
| 20 | 700 | 8D | -27.034 | -27.034 |
| 20 | 700 | global | inf | inf |
| 50 | 700 | 0D | 1.000 | 1.000 |
| 50 | 700 | 1D | -5.897 | -5.897 |
| 50 | 700 | 2D | -12.341 | -15.225 |
| 50 | 700 | 3D | -12.735 | -15.705 |
| 50 | 700 | 4D | -20.772 | -20.772 |
| 50 | 700 | 5D | -20.680 | -20.680 |
| 50 | 700 | 6D | -26.750 | -26.750 |
| 50 | 700 | 8D | -32.517 | -32.517 |
| 50 | 700 | global | -30.592 | -4.833 |
| 100 | 700 | 0D | 1.000 | 1.000 |
| 100 | 700 | 1D | -7.668 | -7.690 |
| 100 | 700 | 2D | -13.851 | -15.372 |
| 100 | 700 | 3D | -14.290 | -15.856 |
| 100 | 700 | 4D | -21.067 | -21.330 |
| 100 | 700 | 5D | -20.974 | -21.236 |
| 100 | 700 | 6D | -26.944 | -27.007 |
| 100 | 700 | 8D | -32.862 | -33.320 |
| 100 | 700 | global | -18.416 | -5.059 |
| 20 | 1000 | 0D | 1.000 | 1.000 |
| 20 | 1000 | 1D | -7.613 | -7.613 |
| 20 | 1000 | 2D | -15.870 | -15.870 |
| 20 | 1000 | 3D | -16.369 | -16.369 |
| 20 | 1000 | 4D | -21.214 | -21.214 |
| 20 | 1000 | 5D | -21.120 | -21.120 |
| 20 | 1000 | 6D | -26.861 | -26.861 |
| 20 | 1000 | 8D | -32.735 | -32.735 |
| 20 | 1000 | global | -23.410 | -3.470 |
| 50 | 1000 | 0D | inf | 1.000 |
| 50 | 1000 | 1D | -7.642 | -7.642 |
| 50 | 1000 | 2D | -15.709 | -15.804 |
| 50 | 1000 | 3D | -16.202 | -16.300 |
| 50 | 1000 | 4D | -21.001 | -21.141 |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1000 | 5D | -20.908 | -21.047 |
| 50 | 1000 | 6D | -26.594 | -26.769 |
| 50 | 1000 | 8D | -32.411 | -32.624 |
| 50 | 1000 | global | inf | -3.611 |
| 100 | 1000 | 0D | inf | 1.000 |
| 100 | 1000 | 1D | -7.670 | -7.670 |
| 100 | 1000 | 2D | -15.972 | -15.972 |
| 100 | 1000 | 3D | -16.474 | -16.474 |
| 100 | 1000 | 4D | -20.892 | -21.258 |
| 100 | 1000 | 5D | -20.799 | -21.164 |
| 100 | 1000 | 6D | -26.457 | -26.916 |
| 100 | 1000 | 8D | -32.246 | -32.802 |
| 100 | 1000 | global | inf | -3.608 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | -6.895 | -7.127 |
| 20 | 1500 | 2D | -13.692 | -14.821 |
| 20 | 1500 | 3D | -14.127 | -15.289 |
| 20 | 1500 | 4D | -18.346 | -19.832 |
| 20 | 1500 | 5D | -18.264 | -19.744 |
| 20 | 1500 | 6D | -23.264 | -25.128 |
| 20 | 1500 | 8D | -28.380 | -30.637 |
| 20 | 1500 | global | inf | -3.936 |
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | -7.026 | -7.950 |
| 50 | 1500 | 2D | -13.936 | -15.495 |
| 50 | 1500 | 3D | -14.378 | -15.983 |
| 50 | 1500 | 4D | -18.667 | -20.720 |
| 50 | 1500 | 5D | -18.584 | -20.628 |
| 50 | 1500 | 6D | -23.667 | -26.241 |
| 50 | 1500 | 8D | -28.867 | -31.985 |
| 50 | 1500 | global | inf | -3.829 |
| 100 | 1500 | 0D | inf | 1.000 |
| 100 | 1500 | 1D | -6.677 | -7.975 |
| 100 | 1500 | 2D | -13.286 | -16.107 |
| 100 | 1500 | 3D | -13.708 | -16.613 |
| 100 | 1500 | 4D | -17.811 | -21.526 |
| 100 | 1500 | 5D | -17.731 | -21.430 |
| 100 | 1500 | 6D | -22.592 | -27.252 |
| 100 | 1500 | 8D | -27.566 | -33.209 |
| 100 | 1500 | global | inf | -3.797 |
| 20 | 2000 | 0D | inf | 1.000 |
| 20 | 2000 | 1D | -7.743 | -7.851 |
| 20 | 2000 | 2D | -15.270 | -15.471 |
| 20 | 2000 | 3D | -15.751 | -15.958 |
| 20 | 2000 | 4D | -20.424 | -20.688 |
| 20 | 2000 | 5D | -20.333 | -20.596 |
| 20 | 2000 | 6D | -25.870 | -26.202 |
| 20 | 2000 | 8D | -31.535 | -31.937 |
| 20 | 2000 | global | inf | -5.109 |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 2000 | 0D | inf | 1.000 |
| 50 | 2000 | 1D | -8.141 | -8.141 |
| 50 | 2000 | 2D | -16.011 | -16.011 |
| 50 | 2000 | 3D | -16.514 | -16.514 |
| 50 | 2000 | 4D | -21.399 | -21.399 |
| 50 | 2000 | 5D | -21.304 | -21.304 |
| 50 | 2000 | 6D | -27.093 | -27.093 |
| 50 | 2000 | 8D | -33.016 | -33.016 |
| 50 | 2000 | global | inf | -4.657 |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | -8.104 | -8.104 |
| 100 | 2000 | 2D | -15.941 | -15.941 |
| 100 | 2000 | 3D | -16.442 | -16.442 |
| 100 | 2000 | 4D | -21.308 | -21.308 |
| 100 | 2000 | 5D | -21.213 | -21.213 |
| 100 | 2000 | 6D | -26.979 | -26.979 |
| 100 | 2000 | 8D | -32.877 | -32.877 |
| 100 | 2000 | global | inf | -4.722 |
| 20 | 3000 | 0D | inf | 1.000 |
| 20 | 3000 | 1D | inf | -6.672 |
| 20 | 3000 | 2D | inf | -13.277 |
| 20 | 3000 | 3D | inf | -13.699 |
| 20 | 3000 | 4D | inf | -17.800 |
| 20 | 3000 | 5D | inf | -17.720 |
| 20 | 3000 | 6D | inf | -22.579 |
| 20 | 3000 | 8D | inf | -27.550 |
| 20 | 3000 | global | inf | -4.228 |
| 50 | 3000 | 0D | inf | 1.000 |
| 50 | 3000 | 1D | inf | -7.624 |
| 50 | 3000 | 2D | inf | -15.049 |
| 50 | 3000 | 3D | inf | -15.523 |
| 50 | 3000 | 4D | inf | -20.132 |
| 50 | 3000 | 5D | inf | -20.043 |
| 50 | 3000 | 6D | inf | -25.504 |
| 50 | 3000 | 8D | inf | -31.092 |
| 50 | 3000 | global | inf | -4.877 |
| 100 | 3000 | 0D | inf | 1.000 |
| 100 | 3000 | 1D | inf | -7.450 |
| 100 | 3000 | 2D | inf | -14.725 |
| 100 | 3000 | 3D | inf | -15.190 |
| 100 | 3000 | 4D | inf | -19.706 |
| 100 | 3000 | 5D | inf | -19.618 |
| 100 | 3000 | 6D | inf | -24.969 |
| 100 | 3000 | 8D | inf | -30.444 |
| 100 | 3000 | global | inf | -4.561 |
| 20 | 4000 | 0D | inf | 1.000 |
| 20 | 4000 | 1D | inf | -7.344 |
| 20 | 4000 | 2D | inf | -14.527 |
| 20 | 4000 | 3D | inf | -14.986 |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 4000 | 4D | inf | -19.445 |
| 20 | 4000 | 5D | inf | -19.358 |
| 20 | 4000 | 6D | inf | -24.642 |
| 20 | 4000 | 8D | inf | -30.049 |
| 20 | 4000 | global | inf | -4.686 |
| 50 | 4000 | 0D | inf | 1.000 |
| 50 | 4000 | 1D | inf | -6.873 |
| 50 | 4000 | 2D | inf | -13.650 |
| 50 | 4000 | 3D | inf | -14.083 |
| 50 | 4000 | 4D | inf | -18.290 |
| 50 | 4000 | 5D | inf | -18.208 |
| 50 | 4000 | 6D | inf | -23.194 |
| 50 | 4000 | 8D | inf | -28.295 |
| 50 | 4000 | global | inf | -4.365 |
| 100 | 4000 | 0D | inf | 1.000 |
| 100 | 4000 | 1D | inf | -7.730 |
| 100 | 4000 | 2D | inf | -15.245 |
| 100 | 4000 | 3D | inf | -15.726 |
| 100 | 4000 | 4D | inf | -20.391 |
| 100 | 4000 | 5D | inf | -20.301 |
| 100 | 4000 | 6D | inf | -25.829 |
| 100 | 4000 | 8D | inf | -31.485 |
| 100 | 4000 | global | inf | -4.949 |

## 2.7 poisson_disk

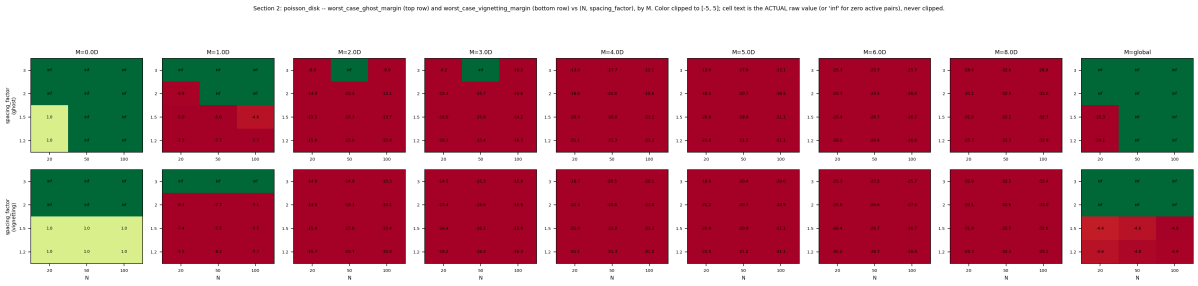


Figure 2.19: poisson_disk: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. ($N$, spacing_factor), by $M$. axes: $N \in \{20, 50, 100\}$.

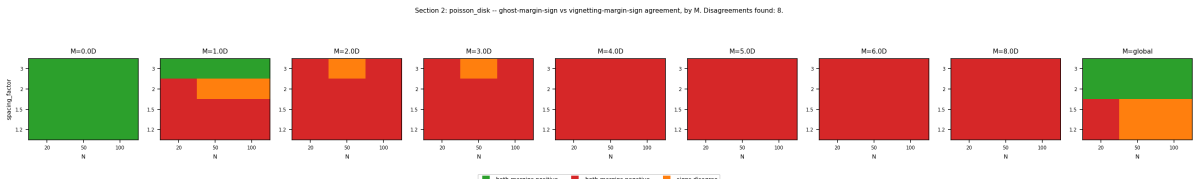


Figure 2.20: poisson_disk: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

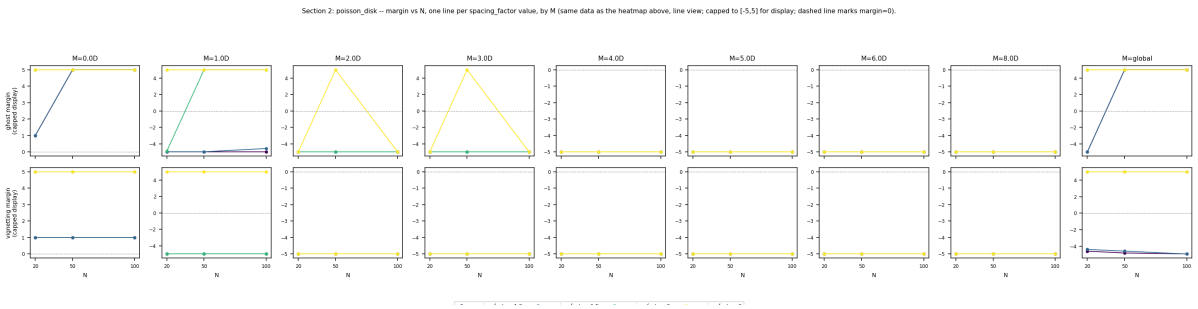



Figure 2.21: poisson_disk: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per spacing_factor value) – a complementary graphical view for reading trends across $N$ directly.

### poisson_disk: raw number list

Table 2.7: poisson_disk: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, spacing_factor, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | spacing_factor | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 1.2 | 0D | 1.000 | 1.000 |
| 20 | 1.2 | 1D | -7.544 | -7.544 |
| 20 | 1.2 | 2D | -15.610 | -15.732 |
| 20 | 1.2 | 3D | -16.101 | -16.227 |
| 20 | 1.2 | 4D | -21.106 | -21.106 |
| 20 | 1.2 | 5D | -21.012 | -21.012 |
| 20 | 1.2 | 6D | -26.188 | -26.188 |
| 20 | 1.2 | 8D | -25.336 | -25.336 |
| 20 | 1.2 | global | -20.074 | -4.584 |
| 50 | 1.2 | 0D | inf | 1.000 |
| 50 | 1.2 | 1D | -7.698 | -8.164 |
| 50 | 1.2 | 2D | -14.958 | -15.722 |
| 50 | 1.2 | 3D | -15.430 | -16.217 |
| 50 | 1.2 | 4D | -21.253 | -21.253 |
| 50 | 1.2 | 5D | -21.159 | -21.159 |
| 50 | 1.2 | 6D | -26.385 | -26.673 |
| 50 | 1.2 | 8D | -32.283 | -32.283 |
| 50 | 1.2 | global | inf | -4.808 |
| 100 | 1.2 | 0D | inf | 1.000 |
| 100 | 1.2 | 1D | -7.674 | -7.674 |
| 100 | 1.2 | 2D | -15.840 | -15.840 |
| 100 | 1.2 | 3D | -16.338 | -16.338 |
| 100 | 1.2 | 4D | -21.234 | -21.234 |
| 100 | 1.2 | 5D | -21.139 | -21.139 |
| 100 | 1.2 | 6D | -26.796 | -26.922 |
| 100 | 1.2 | 8D | -32.879 | -33.130 |
| 100 | 1.2 | global | inf | -4.912 |
| 20 | 1.5 | 0D | 1.000 | 1.000 |
| 20 | 1.5 | 1D | -4.972 | -7.367 |
| 20 | 1.5 | 2D | -15.469 | -15.930 |
| 20 | 1.5 | 3D | -15.956 | -16.430 |
| 20 | 1.5 | 4D | -20.871 | -21.032 |
| 20 | 1.5 | 5D | -20.778 | -20.939 |
| 20 | 1.5 | 6D | -26.356 | -26.356 |
| 20 | 1.5 | 8D | -31.920 | -31.920 |
| 20 | 1.5 | global | -25.343 | -4.375 |
| 50 | 1.5 | 0D | inf | 1.000 |
| 50 | 1.5 | 1D | -4.972 | -7.492 |
| 50 | 1.5 | 2D | -15.299 | -15.584 |
| 50 | 1.5 | 3D | -15.781 | -16.074 |
| 50 | 1.5 | 4D | -20.013 | -21.019 |

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| N | spacing_factor | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1.5 | 5D | -19.924 | -20.926 |
| 50 | 1.5 | 6D | -26.725 | -26.725 |
| 50 | 1.5 | 8D | -32.158 | -32.507 |
| 50 | 1.5 | global | inf | -4.578 |
| 100 | 1.5 | 0D | inf | 1.000 |
| 100 | 1.5 | 1D | -4.577 | -7.527 |
| 100 | 1.5 | 2D | -13.725 | -15.409 |
| 100 | 1.5 | 3D | -14.160 | -15.894 |
| 100 | 1.5 | 4D | -21.174 | -21.174 |
| 100 | 1.5 | 5D | -21.080 | -21.080 |
| 100 | 1.5 | 6D | -26.687 | -26.687 |
| 100 | 1.5 | 8D | -32.656 | -32.808 |
| 100 | 1.5 | global | inf | -4.927 |
| 20 | 2 | 0D | inf | inf |
| 20 | 2 | 1D | -4.899 | -8.085 |
| 20 | 2 | 2D | -14.900 | -14.900 |
| 20 | 2 | 3D | -15.370 | -15.370 |
| 20 | 2 | 4D | -18.631 | -21.313 |
| 20 | 2 | 5D | -18.547 | -21.218 |
| 20 | 2 | 6D | -25.733 | -26.830 |
| 20 | 2 | 8D | -33.081 | -33.081 |
| 20 | 2 | global | inf | inf |
| 50 | 2 | 0D | inf | inf |
| 50 | 2 | 1D | inf | -7.742 |
| 50 | 2 | 2D | -15.185 | -16.053 |
| 50 | 2 | 3D | -15.663 | -16.557 |
| 50 | 2 | 4D | -20.763 | -20.842 |
| 50 | 2 | 5D | -20.670 | -20.749 |
| 50 | 2 | 6D | -24.487 | -26.590 |
| 50 | 2 | 8D | -32.491 | -32.582 |
| 50 | 2 | global | inf | inf |
| 100 | 2 | 0D | inf | inf |
| 100 | 2 | 1D | inf | -7.110 |
| 100 | 2 | 2D | -15.142 | -15.142 |
| 100 | 2 | 3D | -15.619 | -15.619 |
| 100 | 2 | 4D | -18.631 | -20.980 |
| 100 | 2 | 5D | -18.547 | -20.886 |
| 100 | 2 | 6D | -25.984 | -27.380 |
| 100 | 2 | 8D | -32.630 | -33.032 |
| 100 | 2 | global | inf | inf |
| 20 | 3 | 0D | inf | inf |
| 20 | 3 | 1D | inf | inf |
| 20 | 3 | 2D | -8.880 | -14.037 |
| 20 | 3 | 3D | -9.172 | -14.482 |
| 20 | 3 | 4D | -13.048 | -18.683 |
| 20 | 3 | 5D | -12.989 | -18.599 |
| 20 | 3 | 6D | -25.259 | -25.259 |
| 20 | 3 | 8D | -28.812 | -32.885 |
| 20 | 3 | global | inf | inf |

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| N | spacing_factor | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 3 | 0D | inf | inf |
| 50 | 3 | 1D | inf | inf |
| 50 | 3 | 2D | inf | -14.854 |
| 50 | 3 | 3D | inf | -15.322 |
| 50 | 3 | 4D | -17.721 | -20.484 |
| 50 | 3 | 5D | -17.642 | -20.393 |
| 50 | 3 | 6D | -25.729 | -27.163 |
| 50 | 3 | 8D | -32.049 | -32.169 |
| 50 | 3 | global | inf | inf |
| 100 | 3 | 0D | inf | inf |
| 100 | 3 | 1D | inf | inf |
| 100 | 3 | 2D | -9.917 | -15.300 |
| 100 | 3 | 3D | -10.240 | -15.782 |
| 100 | 3 | 4D | -12.118 | -20.059 |
| 100 | 3 | 5D | -12.062 | -19.970 |
| 100 | 3 | 6D | -25.658 | -25.658 |
| 100 | 3 | 8D | -28.812 | -32.379 |
| 100 | 3 | global | inf | inf |

## 2.8 rectangular

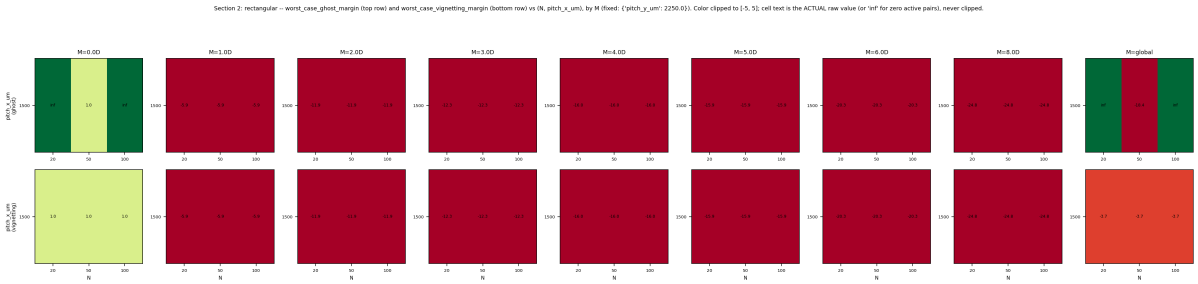



Figure 2.22: rectangular: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \text{pitch_x_um})$, by $M$. pitch_y_um=2250.0 held fixed at its own median tested value; axes: $N \in \{20, 50, 100\}$.

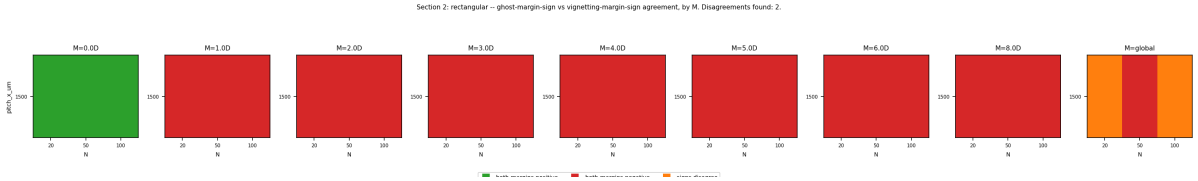



Figure 2.23: rectangular: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

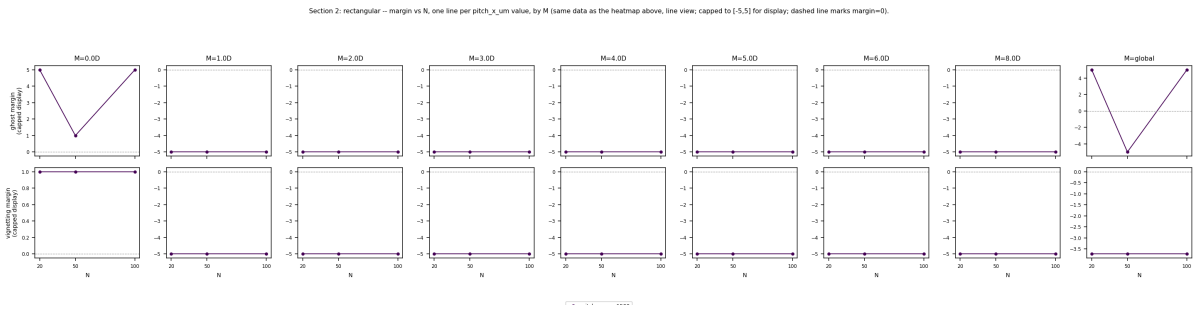



Figure 2.24: rectangular: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per pitch_x_um value) – a complementary graphical view for reading trends across $N$ directly.

### rectangular: raw number list

Table 2.8: rectangular: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, pitch_x_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | pitch_x_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | -5.939 | -5.939 |
| 20 | 1500 | 2D | -11.913 | -11.913 |
| 20 | 1500 | 3D | -12.295 | -12.295 |
| 20 | 1500 | 4D | -16.003 | -16.003 |
| 20 | 1500 | 5D | -15.931 | -15.931 |
| 20 | 1500 | 6D | -20.326 | -20.326 |
| 20 | 1500 | 8D | -24.822 | -24.822 |
| 20 | 1500 | global | inf | -3.729 |
| 50 | 1500 | 0D | 1.000 | 1.000 |
| 50 | 1500 | 1D | -5.939 | -5.939 |
| 50 | 1500 | 2D | -11.913 | -11.913 |
| 50 | 1500 | 3D | -12.295 | -12.295 |
| 50 | 1500 | 4D | -16.003 | -16.003 |
| 50 | 1500 | 5D | -15.931 | -15.931 |
| 50 | 1500 | 6D | -20.326 | -20.326 |
| 50 | 1500 | 8D | -24.822 | -24.822 |
| 50 | 1500 | global | -18.366 | -3.729 |
| 100 | 1500 | 0D | inf | 1.000 |
| 100 | 1500 | 1D | -5.939 | -5.939 |
| 100 | 1500 | 2D | -11.913 | -11.913 |
| 100 | 1500 | 3D | -12.295 | -12.295 |
| 100 | 1500 | 4D | -16.003 | -16.003 |
| 100 | 1500 | 5D | -15.931 | -15.931 |
| 100 | 1500 | 6D | -20.326 | -20.326 |
| 100 | 1500 | 8D | -24.822 | -24.822 |
| 100 | 1500 | global | inf | -3.729 |

## 2.9 ring_offset_concentric

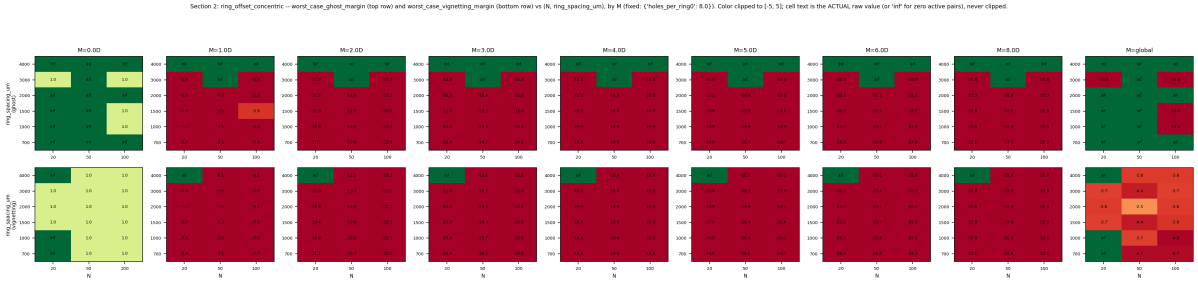


Figure 2.25: ring_offset_concentric: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \text{ring_spacing_um})$, by $M$. holes_per_ring0=8.0 held fixed at its own median tested value; axes: $N \in \{20, 50, 100\}$.

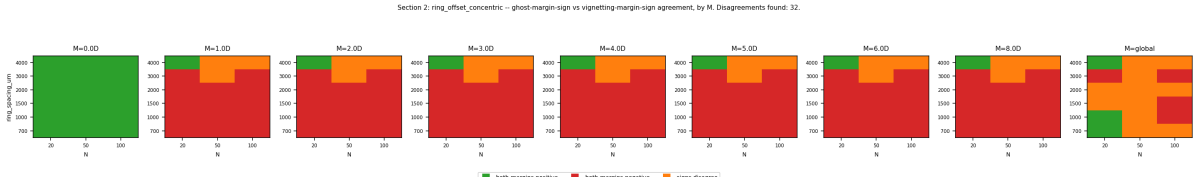


Figure 2.26: ring_offset_concentric: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

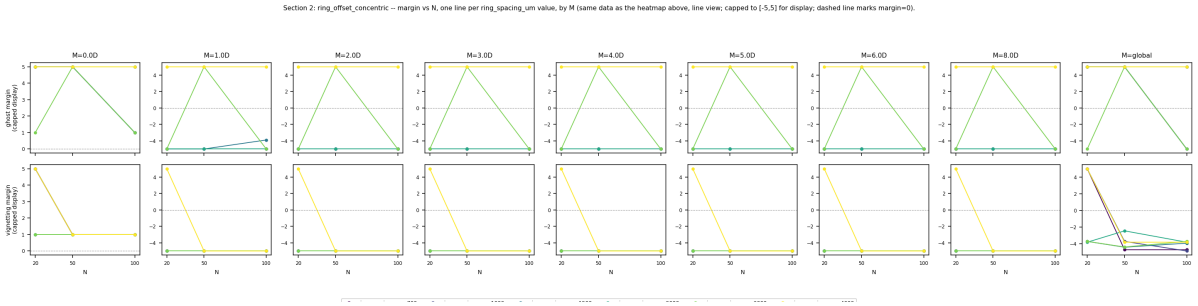



Figure 2.27: ring_offset_concentric: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per ring_spacing_um value) – a complementary graphical view for reading trends across $N$ directly.

### ring_offset_concentric: raw number list

Table 2.9: ring_offset_concentric: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, ring_spacing_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 700 | 0D | inf | inf |
| 20 | 700 | 1D | -5.352 | -5.352 |
| 20 | 700 | 2D | -11.052 | -11.052 |
| 20 | 700 | 3D | -11.408 | -11.408 |
| 20 | 700 | 4D | -14.870 | -14.870 |
| 20 | 700 | 5D | -14.802 | -14.802 |
| 20 | 700 | 6D | -18.904 | -18.904 |
| 20 | 700 | 8D | -23.100 | -23.100 |
| 20 | 700 | global | inf | inf |
| 50 | 700 | 0D | inf | 1.000 |
| 50 | 700 | 1D | -7.079 | -7.079 |
| 50 | 700 | 2D | -13.787 | -15.702 |
| 50 | 700 | 3D | -14.225 | -16.196 |
| 50 | 700 | 4D | -18.796 | -19.615 |
| 50 | 700 | 5D | -18.712 | -19.528 |
| 50 | 700 | 6D | -18.904 | -26.583 |
| 50 | 700 | 8D | -23.100 | -32.399 |
| 50 | 700 | global | inf | -4.733 |
| 100 | 700 | 0D | inf | 1.000 |
| 100 | 700 | 1D | -5.352 | -7.703 |
| 100 | 700 | 2D | -13.787 | -15.542 |
| 100 | 700 | 3D | -14.225 | -16.031 |
| 100 | 700 | 4D | -18.796 | -19.779 |
| 100 | 700 | 5D | -18.712 | -19.691 |
| 100 | 700 | 6D | -24.856 | -26.319 |
| 100 | 700 | 8D | -30.307 | -32.079 |
| 100 | 700 | global | inf | -4.733 |
| 20 | 1000 | 0D | inf | inf |
| 20 | 1000 | 1D | -7.548 | -7.548 |
| 20 | 1000 | 2D | -14.907 | -14.907 |
| 20 | 1000 | 3D | -15.377 | -15.377 |
| 20 | 1000 | 4D | -19.945 | -19.945 |
| 20 | 1000 | 5D | -19.857 | -19.857 |
| 20 | 1000 | 6D | -25.270 | -25.270 |
| 20 | 1000 | 8D | -30.808 | -30.808 |
| 20 | 1000 | global | inf | inf |
| 50 | 1000 | 0D | inf | 1.000 |
| 50 | 1000 | 1D | -5.542 | -5.939 |
| 50 | 1000 | 2D | -14.907 | -15.244 |
| 50 | 1000 | 3D | -15.377 | -15.725 |
| 50 | 1000 | 4D | -19.945 | -20.390 |

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| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1000 | 5D | -19.857 | -20.299 |
| 50 | 1000 | 6D | -25.270 | -25.827 |
| 50 | 1000 | 8D | -30.808 | -31.483 |
| 50 | 1000 | global | inf | -3.729 |
| 100 | 1000 | 0D | 1.000 | 1.000 |
| 100 | 1000 | 1D | -5.542 | -6.995 |
| 100 | 1000 | 2D | -14.907 | -15.070 |
| 100 | 1000 | 3D | -15.377 | -15.545 |
| 100 | 1000 | 4D | -19.945 | -20.160 |
| 100 | 1000 | 5D | -19.857 | -20.071 |
| 100 | 1000 | 6D | -25.270 | -25.539 |
| 100 | 1000 | 8D | -30.808 | -31.135 |
| 100 | 1000 | global | -24.822 | -4.882 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | -5.939 | -5.939 |
| 20 | 1500 | 2D | -11.913 | -13.437 |
| 20 | 1500 | 3D | -12.295 | -13.864 |
| 20 | 1500 | 4D | -16.003 | -18.010 |
| 20 | 1500 | 5D | -15.931 | -17.930 |
| 20 | 1500 | 6D | -20.326 | -22.843 |
| 20 | 1500 | 8D | -24.822 | -27.870 |
| 20 | 1500 | global | inf | -3.729 |
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | -5.939 | -7.325 |
| 50 | 1500 | 2D | -11.913 | -13.825 |
| 50 | 1500 | 3D | -12.295 | -14.263 |
| 50 | 1500 | 4D | -16.003 | -18.521 |
| 50 | 1500 | 5D | -15.931 | -18.438 |
| 50 | 1500 | 6D | -20.326 | -23.483 |
| 50 | 1500 | 8D | -24.822 | -28.645 |
| 50 | 1500 | global | inf | -4.429 |
| 100 | 1500 | 0D | 1.000 | 1.000 |
| 100 | 1500 | 1D | -3.907 | -7.867 |
| 100 | 1500 | 2D | -10.930 | -15.500 |
| 100 | 1500 | 3D | -11.283 | -15.988 |
| 100 | 1500 | 4D | -14.709 | -20.726 |
| 100 | 1500 | 5D | -14.642 | -20.634 |
| 100 | 1500 | 6D | -18.702 | -26.249 |
| 100 | 1500 | 8D | -22.856 | -31.994 |
| 100 | 1500 | global | -24.822 | -3.942 |
| 20 | 2000 | 0D | inf | 1.000 |
| 20 | 2000 | 1D | -5.542 | -7.548 |
| 20 | 2000 | 2D | -11.175 | -14.907 |
| 20 | 2000 | 3D | -11.534 | -15.377 |
| 20 | 2000 | 4D | -15.031 | -19.945 |
| 20 | 2000 | 5D | -14.963 | -19.857 |
| 20 | 2000 | 6D | -19.106 | -25.270 |
| 20 | 2000 | 8D | -23.345 | -30.808 |
| 20 | 2000 | global | inf | -3.825 |

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| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 2000 | 0D | inf | 1.000 |
| 50 | 2000 | 1D | -7.548 | -7.548 |
| 50 | 2000 | 2D | -14.907 | -14.907 |
| 50 | 2000 | 3D | -15.377 | -15.377 |
| 50 | 2000 | 4D | -19.945 | -19.945 |
| 50 | 2000 | 5D | -19.857 | -19.857 |
| 50 | 2000 | 6D | -25.270 | -25.270 |
| 50 | 2000 | 8D | -30.808 | -30.808 |
| 50 | 2000 | global | inf | -2.461 |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | -5.542 | -6.081 |
| 100 | 2000 | 2D | -11.175 | -12.178 |
| 100 | 2000 | 3D | -11.534 | -12.567 |
| 100 | 2000 | 4D | -15.031 | -16.352 |
| 100 | 2000 | 5D | -14.963 | -16.278 |
| 100 | 2000 | 6D | -19.106 | -20.763 |
| 100 | 2000 | 8D | -23.345 | -25.351 |
| 100 | 2000 | global | inf | -3.825 |
| 20 | 3000 | 0D | 1.000 | 1.000 |
| 20 | 3000 | 1D | -5.939 | -5.939 |
| 20 | 3000 | 2D | -11.913 | -11.913 |
| 20 | 3000 | 3D | -12.295 | -12.295 |
| 20 | 3000 | 4D | -16.003 | -16.003 |
| 20 | 3000 | 5D | -15.931 | -15.931 |
| 20 | 3000 | 6D | -20.326 | -20.326 |
| 20 | 3000 | 8D | -24.822 | -24.822 |
| 20 | 3000 | global | -24.822 | -3.729 |
| 50 | 3000 | 0D | inf | 1.000 |
| 50 | 3000 | 1D | inf | -6.926 |
| 50 | 3000 | 2D | inf | -13.748 |
| 50 | 3000 | 3D | inf | -14.185 |
| 50 | 3000 | 4D | inf | -18.420 |
| 50 | 3000 | 5D | inf | -18.338 |
| 50 | 3000 | 6D | inf | -23.357 |
| 50 | 3000 | 8D | inf | -28.492 |
| 50 | 3000 | global | inf | -4.401 |
| 100 | 3000 | 0D | 1.000 | 1.000 |
| 100 | 3000 | 1D | -5.939 | -5.939 |
| 100 | 3000 | 2D | -11.913 | -11.913 |
| 100 | 3000 | 3D | -12.295 | -12.295 |
| 100 | 3000 | 4D | -16.003 | -16.003 |
| 100 | 3000 | 5D | -15.931 | -15.931 |
| 100 | 3000 | 6D | -20.326 | -20.326 |
| 100 | 3000 | 8D | -24.822 | -24.822 |
| 100 | 3000 | global | -24.822 | -3.729 |
| 20 | 4000 | 0D | inf | inf |
| 20 | 4000 | 1D | inf | inf |
| 20 | 4000 | 2D | inf | inf |
| 20 | 4000 | 3D | inf | inf |

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| N | ring_spacing_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 4000 | 4D | inf | inf |
| 20 | 4000 | 5D | inf | inf |
| 20 | 4000 | 6D | inf | inf |
| 20 | 4000 | 8D | inf | inf |
| 20 | 4000 | global | inf | inf |
| 50 | 4000 | 0D | inf | 1.000 |
| 50 | 4000 | 1D | inf | -6.081 |
| 50 | 4000 | 2D | inf | -12.178 |
| 50 | 4000 | 3D | inf | -12.567 |
| 50 | 4000 | 4D | inf | -16.352 |
| 50 | 4000 | 5D | inf | -16.278 |
| 50 | 4000 | 6D | inf | -20.763 |
| 50 | 4000 | 8D | inf | -25.351 |
| 50 | 4000 | global | inf | -3.825 |
| 100 | 4000 | 0D | inf | 1.000 |
| 100 | 4000 | 1D | inf | -6.081 |
| 100 | 4000 | 2D | inf | -12.178 |
| 100 | 4000 | 3D | inf | -12.567 |
| 100 | 4000 | 4D | inf | -16.352 |
| 100 | 4000 | 5D | inf | -16.278 |
| 100 | 4000 | 6D | inf | -20.763 |
| 100 | 4000 | 8D | inf | -25.351 |
| 100 | 4000 | global | inf | -3.825 |

## 2.10 square

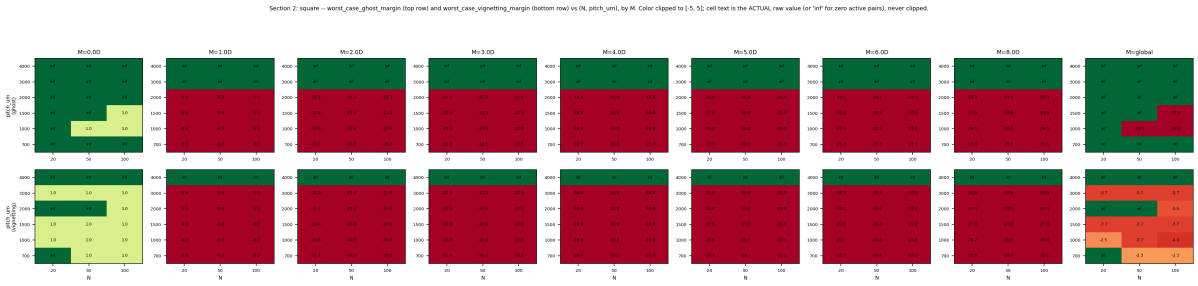



Figure 2.28: square: worst_case_ghost_margin (top row) and worst_case_vignetting_margin (bottom row) vs. $(N, \text{pitch_um})$, by $M$. axes: $N \in \{20, 50, 100\}$.

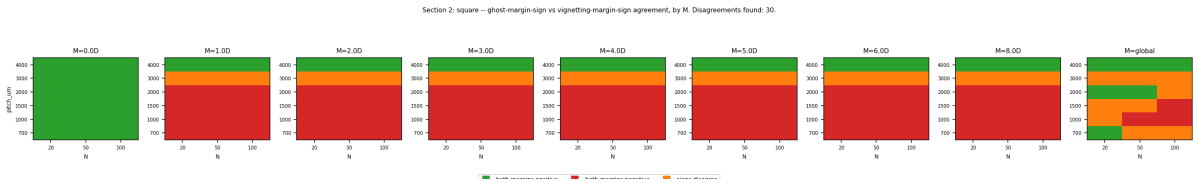



Figure 2.29: square: ghost-margin-sign vs. vignetting-margin-sign agreement, by $M$ (see the figure's own title for the exact disagreement count).

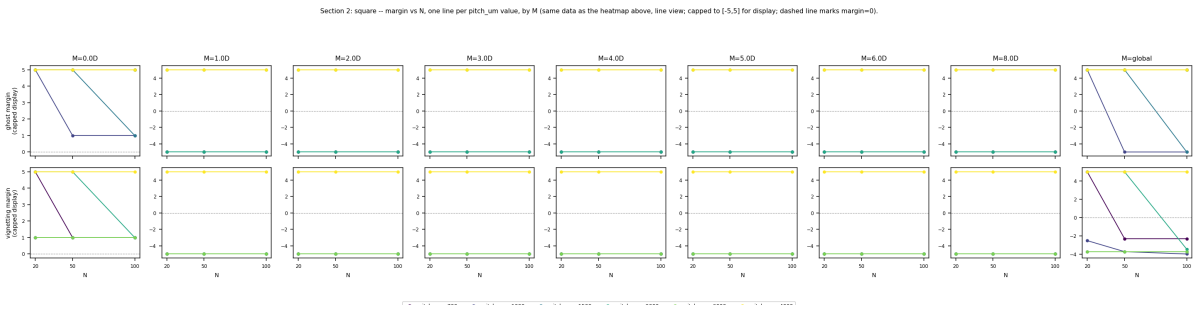



Figure 2.30: square: same margin data as the heatmap above, shown as line plots (margin vs. $N$, one line per pitch_um value) – a complementary graphical view for reading trends across $N$ directly.

### square: raw number list

Table 2.10: square: raw worst_case_ghost_margin and worst_case_vignetting_margin values behind the heatmap and line plot above – one row per tested $(N, pitch_um, M)$ point. 'inf' = zero active hole pairs under the pupil at that point (structurally safe, undisplayable on a linear scale).

| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 700 | 0D | inf | inf |
| 20 | 700 | 1D | -5.477 | -5.477 |
| 20 | 700 | 2D | -11.052 | -11.052 |
| 20 | 700 | 3D | -11.408 | -11.408 |
| 20 | 700 | 4D | -16.743 | -16.743 |
| 20 | 700 | 5D | -16.668 | -16.668 |
| 20 | 700 | 6D | -21.253 | -21.253 |
| 20 | 700 | 8D | -25.945 | -25.945 |
| 20 | 700 | global | inf | inf |
| 50 | 700 | 0D | inf | 1.000 |
| 50 | 700 | 1D | -8.159 | -8.159 |
| 50 | 700 | 2D | -16.044 | -16.044 |
| 50 | 700 | 3D | -16.548 | -16.548 |
| 50 | 700 | 4D | -18.837 | -19.230 |
| 50 | 700 | 5D | -18.753 | -19.144 |
| 50 | 700 | 6D | -23.880 | -24.373 |
| 50 | 700 | 8D | -29.125 | -29.722 |
| 50 | 700 | global | inf | -2.310 |
| 100 | 700 | 0D | inf | 1.000 |
| 100 | 700 | 1D | -8.159 | -8.159 |
| 100 | 700 | 2D | -16.044 | -16.044 |
| 100 | 700 | 3D | -16.548 | -16.548 |
| 100 | 700 | 4D | -21.443 | -21.443 |
| 100 | 700 | 5D | -21.348 | -21.348 |
| 100 | 700 | 6D | -27.148 | -27.148 |
| 100 | 700 | 8D | -33.083 | -33.083 |
| 100 | 700 | global | inf | -2.310 |
| 20 | 1000 | 0D | inf | 1.000 |
| 20 | 1000 | 1D | -8.252 | -8.252 |
| 20 | 1000 | 2D | -12.612 | -12.612 |
| 20 | 1000 | 3D | -13.014 | -13.014 |
| 20 | 1000 | 4D | -16.923 | -16.923 |
| 20 | 1000 | 5D | -16.847 | -16.847 |
| 20 | 1000 | 6D | -21.479 | -21.479 |
| 20 | 1000 | 8D | -26.218 | -26.218 |
| 20 | 1000 | global | inf | -2.524 |
| 50 | 1000 | 0D | 1.000 | 1.000 |
| 50 | 1000 | 1D | -8.252 | -8.252 |
| 50 | 1000 | 2D | -12.612 | -14.520 |
| 50 | 1000 | 3D | -13.014 | -14.978 |
| 50 | 1000 | 4D | -16.923 | -19.435 |

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| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 1000 | 5D | -16.847 | -19.349 |
| 50 | 1000 | 6D | -21.479 | -24.630 |
| 50 | 1000 | 8D | -26.218 | -30.034 |
| 50 | 1000 | global | -18.246 | -3.729 |
| 100 | 1000 | 0D | 1.000 | 1.000 |
| 100 | 1000 | 1D | -6.315 | -6.315 |
| 100 | 1000 | 2D | -12.612 | -14.520 |
| 100 | 1000 | 3D | -13.014 | -14.978 |
| 100 | 1000 | 4D | -16.923 | -19.435 |
| 100 | 1000 | 5D | -16.847 | -19.349 |
| 100 | 1000 | 6D | -21.479 | -24.630 |
| 100 | 1000 | 8D | -26.218 | -30.034 |
| 100 | 1000 | global | -18.246 | -3.984 |
| 20 | 1500 | 0D | inf | 1.000 |
| 20 | 1500 | 1D | -6.758 | -6.758 |
| 20 | 1500 | 2D | -13.437 | -13.437 |
| 20 | 1500 | 3D | -13.864 | -13.864 |
| 20 | 1500 | 4D | -18.010 | -18.010 |
| 20 | 1500 | 5D | -17.930 | -17.930 |
| 20 | 1500 | 6D | -22.843 | -22.843 |
| 20 | 1500 | 8D | -27.870 | -27.870 |
| 20 | 1500 | global | inf | -3.729 |
| 50 | 1500 | 0D | inf | 1.000 |
| 50 | 1500 | 1D | -5.939 | -6.758 |
| 50 | 1500 | 2D | -11.913 | -13.437 |
| 50 | 1500 | 3D | -12.295 | -13.864 |
| 50 | 1500 | 4D | -16.003 | -18.010 |
| 50 | 1500 | 5D | -15.931 | -17.930 |
| 50 | 1500 | 6D | -20.326 | -22.843 |
| 50 | 1500 | 8D | -24.822 | -27.870 |
| 50 | 1500 | global | inf | -3.729 |
| 100 | 1500 | 0D | 1.000 | 1.000 |
| 100 | 1500 | 1D | -5.939 | -6.758 |
| 100 | 1500 | 2D | -11.913 | -13.437 |
| 100 | 1500 | 3D | -12.295 | -13.864 |
| 100 | 1500 | 4D | -16.003 | -18.010 |
| 100 | 1500 | 5D | -15.931 | -17.930 |
| 100 | 1500 | 6D | -20.326 | -22.843 |
| 100 | 1500 | 8D | -24.822 | -27.870 |
| 100 | 1500 | global | -17.259 | -3.729 |
| 20 | 2000 | 0D | inf | inf |
| 20 | 2000 | 1D | -5.542 | -5.542 |
| 20 | 2000 | 2D | -11.175 | -11.175 |
| 20 | 2000 | 3D | -11.534 | -11.534 |
| 20 | 2000 | 4D | -15.031 | -15.031 |
| 20 | 2000 | 5D | -14.963 | -14.963 |
| 20 | 2000 | 6D | -19.106 | -19.106 |
| 20 | 2000 | 8D | -23.345 | -23.345 |
| 20 | 2000 | global | inf | inf |

*continued on next page*

| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 50 | 2000 | 0D | inf | inf |
| 50 | 2000 | 1D | -5.542 | -5.542 |
| 50 | 2000 | 2D | -11.175 | -11.175 |
| 50 | 2000 | 3D | -11.534 | -11.534 |
| 50 | 2000 | 4D | -15.031 | -15.031 |
| 50 | 2000 | 5D | -14.963 | -14.963 |
| 50 | 2000 | 6D | -19.106 | -19.106 |
| 50 | 2000 | 8D | -23.345 | -23.345 |
| 50 | 2000 | global | inf | inf |
| 100 | 2000 | 0D | inf | 1.000 |
| 100 | 2000 | 1D | -5.542 | -5.542 |
| 100 | 2000 | 2D | -11.175 | -11.175 |
| 100 | 2000 | 3D | -11.534 | -11.534 |
| 100 | 2000 | 4D | -15.031 | -15.031 |
| 100 | 2000 | 5D | -14.963 | -14.963 |
| 100 | 2000 | 6D | -19.106 | -19.106 |
| 100 | 2000 | 8D | -23.345 | -23.345 |
| 100 | 2000 | global | inf | -3.458 |
| 20 | 3000 | 0D | inf | 1.000 |
| 20 | 3000 | 1D | inf | -5.939 |
| 20 | 3000 | 2D | inf | -11.913 |
| 20 | 3000 | 3D | inf | -12.295 |
| 20 | 3000 | 4D | inf | -16.003 |
| 20 | 3000 | 5D | inf | -15.931 |
| 20 | 3000 | 6D | inf | -20.326 |
| 20 | 3000 | 8D | inf | -24.822 |
| 20 | 3000 | global | inf | -3.729 |
| 50 | 3000 | 0D | inf | 1.000 |
| 50 | 3000 | 1D | inf | -5.939 |
| 50 | 3000 | 2D | inf | -11.913 |
| 50 | 3000 | 3D | inf | -12.295 |
| 50 | 3000 | 4D | inf | -16.003 |
| 50 | 3000 | 5D | inf | -15.931 |
| 50 | 3000 | 6D | inf | -20.326 |
| 50 | 3000 | 8D | inf | -24.822 |
| 50 | 3000 | global | inf | -3.729 |
| 100 | 3000 | 0D | inf | 1.000 |
| 100 | 3000 | 1D | inf | -5.939 |
| 100 | 3000 | 2D | inf | -11.913 |
| 100 | 3000 | 3D | inf | -12.295 |
| 100 | 3000 | 4D | inf | -16.003 |
| 100 | 3000 | 5D | inf | -15.931 |
| 100 | 3000 | 6D | inf | -20.326 |
| 100 | 3000 | 8D | inf | -24.822 |
| 100 | 3000 | global | inf | -3.729 |
| 20 | 4000 | 0D | inf | inf |
| 20 | 4000 | 1D | inf | inf |
| 20 | 4000 | 2D | inf | inf |
| 20 | 4000 | 3D | inf | inf |

*continued on next page*

| N | pitch_um | M | ghost_margin | vignetting_margin |
|---|---|---|---|---|
| 20 | 4000 | 4D | inf | inf |
| 20 | 4000 | 5D | inf | inf |
| 20 | 4000 | 6D | inf | inf |
| 20 | 4000 | 8D | inf | inf |
| 20 | 4000 | global | inf | inf |
| 50 | 4000 | 0D | inf | inf |
| 50 | 4000 | 1D | inf | inf |
| 50 | 4000 | 2D | inf | inf |
| 50 | 4000 | 3D | inf | inf |
| 50 | 4000 | 4D | inf | inf |
| 50 | 4000 | 5D | inf | inf |
| 50 | 4000 | 6D | inf | inf |
| 50 | 4000 | 8D | inf | inf |
| 50 | 4000 | global | inf | inf |
| 100 | 4000 | 0D | inf | inf |
| 100 | 4000 | 1D | inf | inf |
| 100 | 4000 | 2D | inf | inf |
| 100 | 4000 | 3D | inf | inf |
| 100 | 4000 | 4D | inf | inf |
| 100 | 4000 | 5D | inf | inf |
| 100 | 4000 | 6D | inf | inf |
| 100 | 4000 | 8D | inf | inf |
| 100 | 4000 | global | inf | inf |

## Chapter 3

# Seed Variance (poisson_disk, jittered_lattice)

For each $(N, \text{spacing})$ combination actually swept, the distribution of `worst_case_ghost_margin` across all 8 tested seeds, held at a single representative $M = 4$D (chosen so the spread shown is seed-driven only – pooling every $M$ into one violin was tried first and rejected, since cross-$M$ variation is far larger than seed variation and would dominate the plot; see the figure-generation script's own documented reasoning). A companion scatter shows seed-to-seed standard deviation vs. the mean margin at each $(N, \text{spacing}, M)$ point (all $M$ shown here, each point's own std computed strictly across seeds at that one $M$) – no threshold is drawn on it; that is a later step.

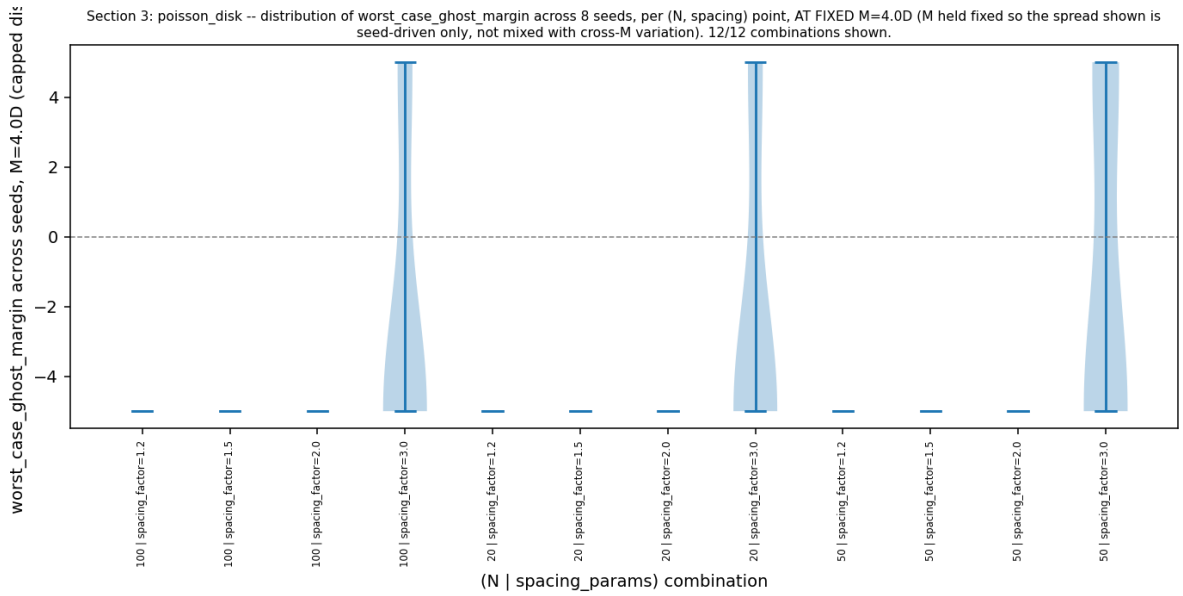



Figure 3.1: poisson_disk: margin distribution across 8 seeds, $M = 4$D, per tested $(N, \text{spacing})$ combination.

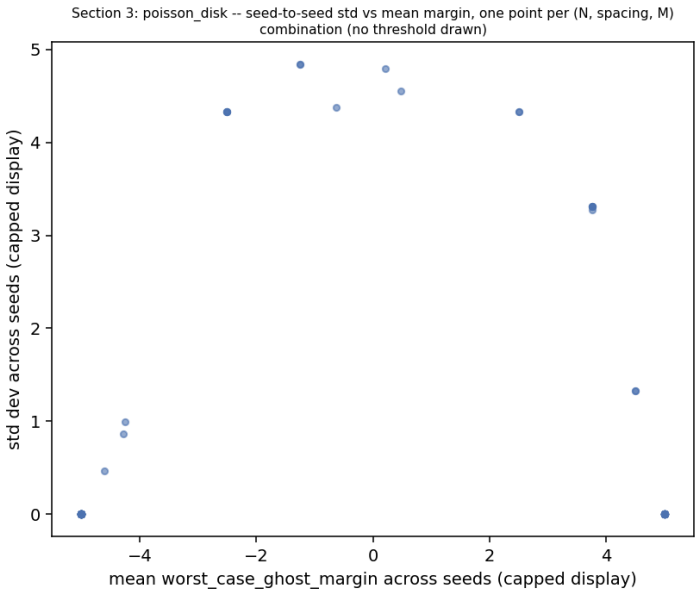



Figure 3.2: poisson_disk: seed-to-seed std vs. mean margin, one point per $(N, \text{spacing}, M)$.

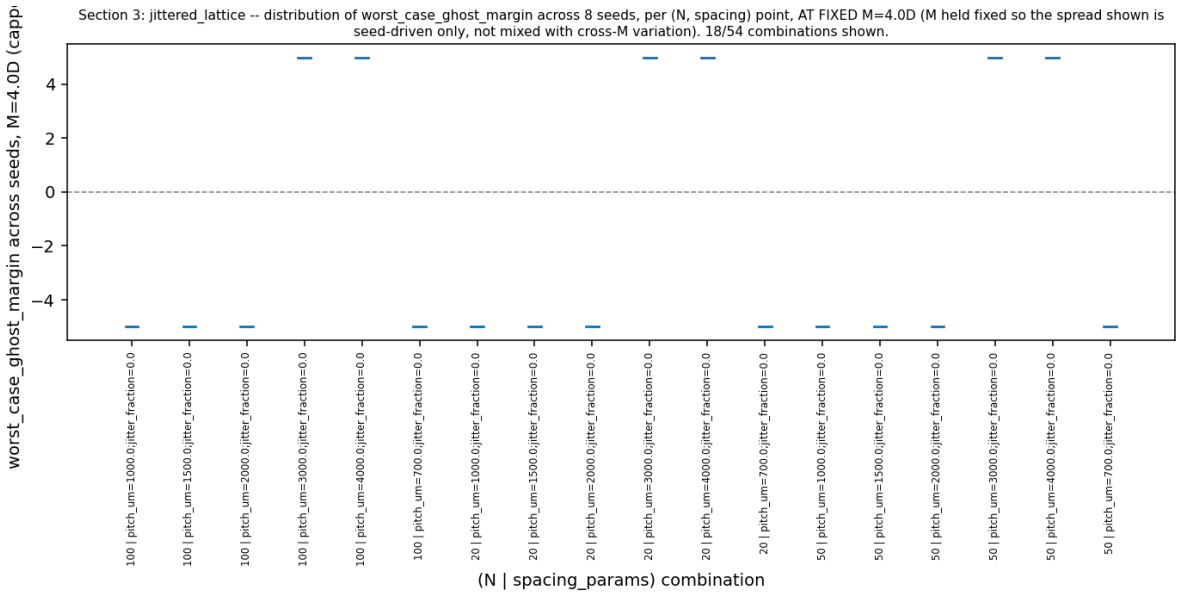



Figure 3.3: jittered_lattice: margin distribution across 8 seeds, $M = 4$D, per tested $(N, \text{spacing})$ combination.

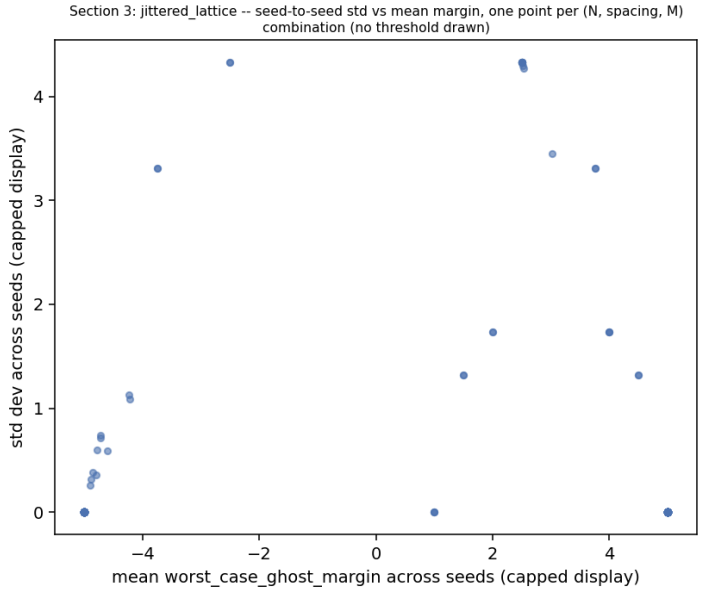



Figure 3.4: jittered_lattice: seed-to-seed std vs. mean margin, one point per $(N, \text{spacing}, M)$.

Chapter 4

# Density vs. Safety

All structurally-safe candidates (`worst_case_ghost_margin` $> 0$ **and** `worst_case_vignetting_margin` $> 0$, read directly from the CSV, not from any gating script), across all 10 families, packing density on the $x$-axis vs. ghost margin on the $y$-axis, colored by family. Shown for $M = 4$D and $M =$ global only (two representative cases, not the full $M$ grid, to keep this chapter to two figures) – $M = 4$D as a representative mid-myopia single-$M$ case, $M =$ global as the cross-$M$ worst-case scope.

**Density computation, stated explicitly**: packing density = `N_actual` / (area of the convex hull of the generated hole-center coordinates), in holes per $\mu$m$^2$. The SAME convex-hull-area definition is applied uniformly to all 10 families (rather than a different closed-form cell-area formula per lattice type), so density values are directly comparable on one axis; irregular families (spirals, rings, stochastic patterns) have no closed-form cell area the way a lattice pitch does, so this is the one measure that applies to all of them without a family-specific exception. Coordinates are regenerated at each plotted point's own exact $(N, \text{spacing}[, \text{seed}])$ to compute this – not estimated.

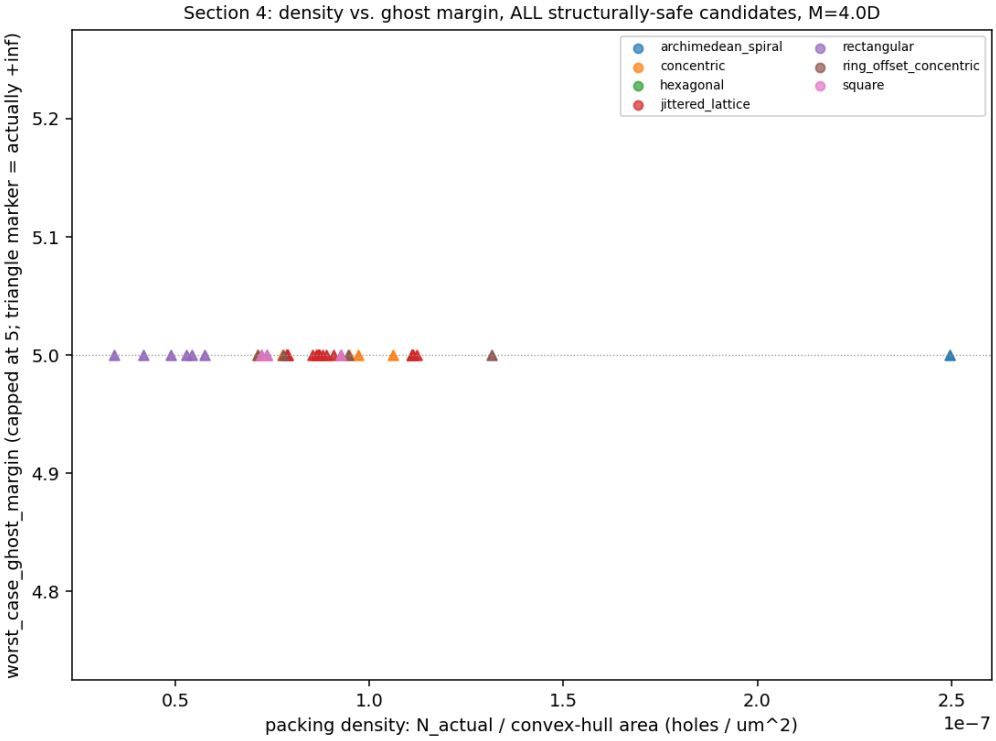



Figure 4.1: Density vs. ghost margin, all structurally-safe candidates, $M = 4$D.

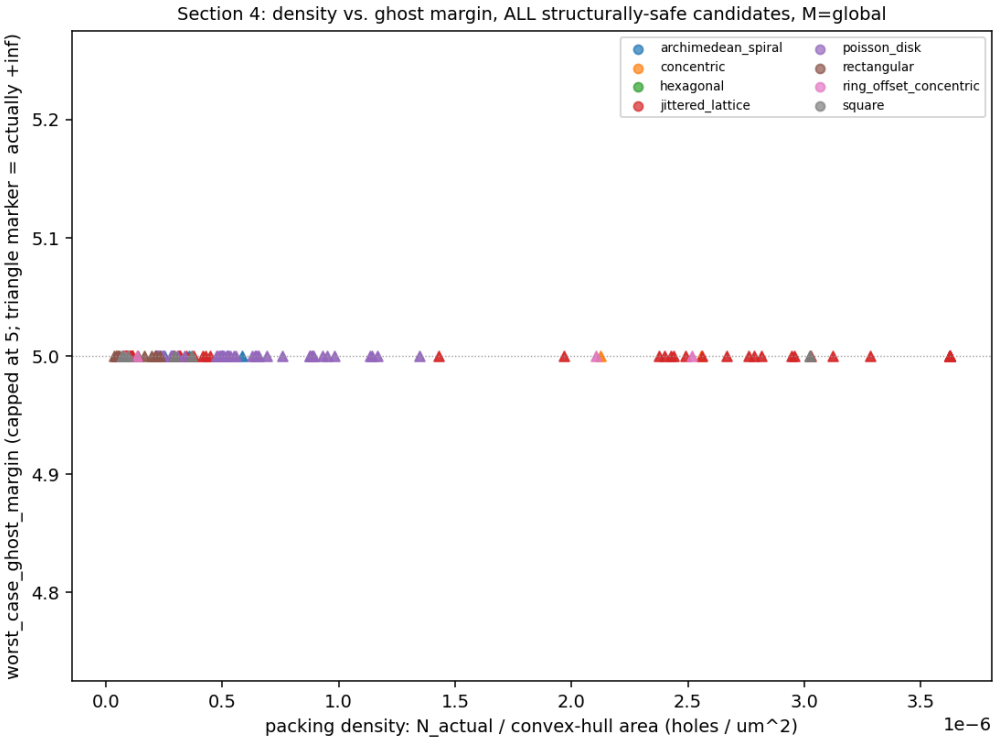



Figure 4.2: Density vs. ghost margin, all structurally-safe candidates, $M$ = global.

## Chapter 5

# Cross-Family Summary Table

One row per family, numbers read directly from `sweep_results.csv` and `family_ranking.csv` by `generate_exploration_figures.py` (`section5_summary_table()`), not computed by hand or typed into this document. `best_finite_ghost_margin` excludes $+\infty$ cells, reported separately as `n_inf_margin_rows`; seed-std columns apply only to the two stochastic families and are computed across *every* tested point (not only the ones that ended up structurally safe overall – restricting to safe-only was tried first and silently excluded exactly the highest-variance points, since a point with a very negative worst-case seed is, by definition, not in the safe set at all).

| family | N_cand. | N_safe | best_finite | N_inf | mean_std | max_std |
|---|---|---|---|---|---|---|
| archimedean_spiral | 15 | 26 | -3.337 | 66 | – | – |
| concentric | 36 | 78 | 1.000 | 102 | – | – |
| fibonacci_spiral | 15 | 15 | 1.000 | 54 | – | – |
| golden_angle | 15 | 15 | 1.000 | 54 | – | – |
| hexagonal | 18 | 43 | 1.000 | 76 | – | – |
| jittered_lattice | 432 | 736 | 1.000 | 1740 | 0.182 | 4.330 |
| poisson_disk | 96 | 180 | 1.000 | 302 | 0.656 | 4.841 |
| rectangular | 54 | 135 | 1.000 | 226 | – | – |
| ring_offset_concentric | 36 | 70 | 1.000 | 106 | – | – |
| square | 18 | 45 | 1.000 | 72 | – | – |

Table 5.1: N_cand. = number of distinct $(N, \text{spacing}[, \text{seed}])$ points tested; N_safe = rows with `structurally_safe=True`; best_finite = largest `worst_case_ghost_margin` excluding $+\infty$; N_inf = count of $+\infty$-margin rows; mean/max_std = seed-to-seed std of `worst_case_ghost_margin` (poisson_disk/jittered_lattice only, "–" otherwise).